

**48NGU Series B Ultra Low NOx
Performance™ 15.2 SEER2 2-Stage Packaged Air
Conditioner and Single Stage Gas Furnace System
with Puron Advance™ (R-454B) Refrigerant
Single and Three Phase
2 to 5 Nominal Tons (Sizes 24-60)**



Product Data

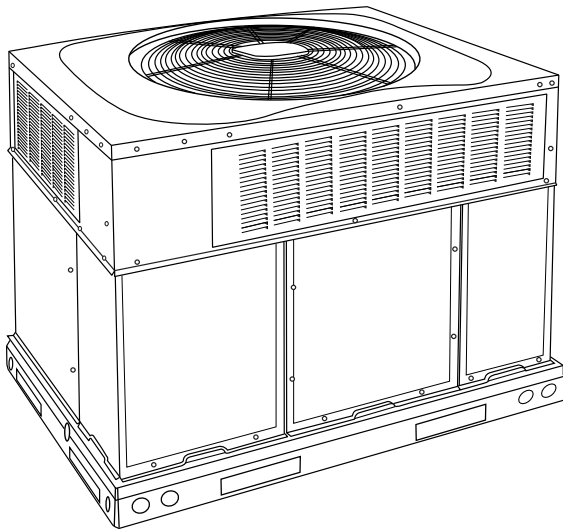


Fig. 1 – Unit 48NG

A09033

Single-Packaged Products with Energy-Saving Features and Puron Advance (R-454B) refrigerant.

- Up to 16.0 SEER2
- 11.5 EER2
- 81% AFUE (Single and three phase models)
- Meets Energy Star requirements
- Direct Spark Ignition
- Factory-Installed TXV
- Multi-speed ECM Blower Motor-Standard
- Sound Levels as low as 68 dBA
- Two Stage Cooling
- Dehumidification Feature
- Refrigerant leak detection dissipation system for added safety

Features/Benefits

One-piece heating and cooling units with low sound levels, easy installation, low maintenance, and dependable performance.

Puron Advance (R-454B) is Carrier's latest choice of refrigerant to help meet the 2025 GWP requirement. This unit is designed and tested with Puron Advance (R-454B) and contains a dissipation system.

Easy Installation

Factory-assembled package is a compact, fully self-contained, combination gas heating/electric cooling unit that is prewired, pre-piped, and pre-charged for minimum installation expense. These units are available in a variety of standard and optional heating/cooling size combinations with voltage options to meet residential and light commercial requirements. Units are lightweight and install easily on a rooftop or at ground level. The high tech composite base eliminates rust problems associated with ground level applications.

Innovative Unit Base Design

On the inside a high-tech composite material will not rust and incorporates a sloped drain pan which improves drainage and helps inhibit mold, algae and bacterial growth. On the outside metal base rails provide added stability as well as easier handling and rigging.

Convertible Duct Configuration

Unit is designed for use in either downflow or horizontal applications. Each unit is converted from horizontal to downflow and includes two horizontal duct covers. Downflow operation is provided in the field to allow vertical ductwork connections. The basepan seals on the bottom openings to ensure a positive seal in the vertical airflow mode.

Efficient Operation

High-efficiency Design offers SEER2 (Seasonal Energy Efficiency Ratios²) of up to 16.0 11.5 EER2, and AFUE (Annual Fuel Utilization Efficiency) ratings as high as 81%.

Energy-saving, Direct Spark Ignition saves gas by operating only when the room thermostat calls for heating. Standard units are furnished with natural gas controls.

Ultra Low NOx units meet South Coast Air Quality Management District (SCAQMD) and San Joaquin Air Pollution Control District (SJAPCD) are regulating a reduction of 65%, down to no more than 14ng/J of NOx emissions.

Durable, dependable components

Compressors have two stages of cooling and are designed for high efficiency. Each compressor is hermetically sealed against contamination to help promote longer life and dependable operation. Each compressor also has vibration isolation to provide quieter operation. All compressors have internal high pressure and overcurrent protection.

Turbo-tubular™ Heat Exchangers are constructed of stainless steel for corrosion resistance and optimum heat transfer for improved efficiency. The tubular design permits hot gases to make multiple passes across the path of the supply air.

In addition, dimples located on the heat exchanger walls force the hot gases to stay in close contact with the walls, improving heat transfer.

Multi-speed ECM Blower Motor is standard on all models.

Direct-drive PSC (Permanent Split Capacitor) condenser-fan motors are designed to help reduce energy consumption and provide for cooling operation down to 40°F (4.4°C) outdoor temperature. Low ambient kit is available as a field-installed accessory.

Thermostatic Expansion Valve - A hard shutoff, balance port TXV maintains a constant superheat at the evaporator exit (cooling cycle) resulting in higher overall system efficiency.

Refrigerant System is designed to provide dependability. Liquid filter driers are used to promote clean, unrestricted operation. Each unit leaves the factory with a full refrigerant charge. Refrigerant service connections

make checking operating pressures easier. Factory-installed leak detection dissipation system for added safety.

High and Low Pressure Switches provide added reliability for the compressor.

Indoor and Outdoor Coils are computer-designed for optimum heat transfer and efficiency. The indoor coil is fabricated from tin-plated copper tube and aluminum fins and is located inside the unit for protection against damage. The outdoor coil is internally mounted on the top tier of the unit.

Low Sound Ratings ensure a quiet indoor and outdoor environment with sound ratings as low as 68dBA.

Dehumidification Feature

This unit has independent fan speeds for low stage cooling and high stage cooling. In addition, 208/230 VAC models have the field-selectable capability to run enhanced dehumidification ('DHUM') speed on each stage cooling (as low as 320CFM per ton). The dehumidification control must open the control circuit on humidity rise above the dehumidification set point.

Heating

- Reliable direct spark ignition system
- Inducer motors with ball bearings

Easy to Service Cabinets provide easy 3-panel accessibility to serviceable components during maintenance and installation. The basepan with integrated drain pan provides easy ground level installation with mounting pad. A nesting feature ensures a positive basepan to roof curb seal when the unit is roof mounted. A convenient 3/4-in. (19.05 mm) wide perimeter flange makes frame mounting on a rooftop easy.

Standard Horizontal Metal Duct Covers with insulation come with the unit and cover the horizontal duct openings. These can be left in place if the units are converted to downflow.

Integrated Gas Control (IGC) Board provides safe and efficient control of heating and simplifies trouble-shooting through its built-in diagnostic function.

Cabinets are constructed of heavyduty, phosphated, zinc-coated prepainted steel capable of withstanding 500 hours in salt spray. Interior surfaces of the evaporator/heat exchanger compartment are insulated with foil-faced insulation, which keeps the conditioned air from being affected by the outdoor ambient temperature and provides improved indoor air quality. (Conforms to American Society of Heating, Refrigeration and Air Conditioning Engineers 62.2.) The sloped drain pan minimizes standing water in the drain. An external drain is provided.

Louvered grille provides hail and vandalism protection for the coil.

Limited Warranty*

- Default 5-year parts limited warranty
 - 10-year parts limited warranty with timely registration. Equipment must be registered within 90 days of original installation, except in jurisdictions where warranty benefits cannot be conditioned on registration.

* Applies to original purchaser/homeowner and 5 years to subsequent owners.

- Default 5-year on compressor limited warranty
 - 10-year on compressor limited warranty with timely registration. Equipment must be registered within 90 days of original installation, except in jurisdictions where warranty benefits cannot be conditioned on registration.

* Applies to original purchaser/homeowner and 5 years to subsequent owners.

- Default 20-year stainless steel heat exchanger limited warranty
 - Life stainless steel limited warranty with timely registration. Equipment must be registered within 90 days of original installation, except in jurisdictions where warranty benefits cannot be conditioned on registration.

* Applies to original purchaser/homeowner and 5 years to subsequent owners. See warranty certificate for complete details and restrictions.

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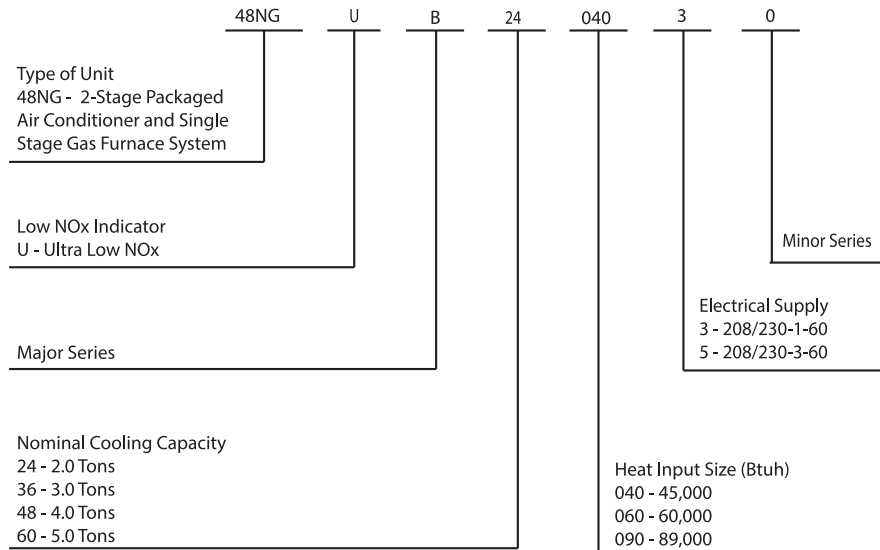
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Model Number Nomenclature



For California Residents:

For installation in SCAQMD only: The Ultra Low NOx furnace does meet the SCAQMD Rule 1111 14ng/J NOx emission limit.

SAP ORDERING NO.	NOMINAL COOLING CAPACITY (Btuh)	NOMINAL HEATING INPUT (Btuh)	VOLTS-PHASE (60 HZ)	APPROX SHIP WT (LB)
48NGUB240403	24,000	45,000	208/230-1	352
48NGUB240603	24,000	60,000	208/230-1	352
48NGUB360603	36,000	60,000	208/230-1	455
48NGUB360605	36,000	60,000	208/230-3	455
48NGUB480903	48,000	89,000	208/230-1	500
48NGUB480905	48,000	89,000	208/230-3	500
48NGUB600903	60,000	89,000	208/230-1	520
48NGUB600905	60,000	89,000	208/230-3	520

AHRI* Capacities

Cooling Capacities and Efficiencies

UNIT SIZE	NOMINAL TONS	STANDARD CFM (High / Low Stage)	COOLING CAPACITY - Btuh (High Stage)	EER2	SEER2
24	2	800 / 600	23200	11.5	15.2
36	3	1200 / 900	34200	11.5	15.2
48	4	1600 / 1200	47000	12.0	16.0
60	5	1750 / 1200	56000	11.5	15.2

LEGEND

dB-Sound Levels (decibels)

db—Dry Bulb

SEER—Seasonal Energy Efficiency Ratio

wb—Wet Bulb

COP—Coefficient of Performance

* Air Conditioning, Heating & Refrigeration Institute.

**At “A” conditions-80°F (26.7°C) indoor db/67°F (19.4°C) indoor wb & 95°F (35°C) outdoor db.

† Rated in accordance with U.S. Government DOE Department of Energy) test procedures and/or AHRI Standards 210/240.

Notes:

1. Ratings are net values, reflecting the effects of circulating fan heat.

Ratings are based on:

Cooling Standard: 80°F (26.7°) db, 67°F wb (19.4°C) indoor entering-air temperature and 95°F db (35°C) outdoor entering-air temperature.

2. Before purchasing this appliance, read important energy cost and efficiency information available from AHRIdirectory.org.

Heating Capacities and Efficiencies

Gas Heating Capacities and Efficiencies, Single and Three Phase Models

Unit Size	Heating Input (Btuh)	Output Capacity (Btuh)	Temperature Rise Range °F (°C)	AFUE (%)
24040	45,000	37,000	25-55 (14-31)	81.0
24060 36060	60,000	49,000	25-55 (14-31)	81.0
48090 60090	89,000	73,000	35-65 (19-36)	81.0

LEGEND

AFUE - Annual Fuel Utilization Efficiency

NOTE: Before purchasing this appliance, read important energy cost and efficiency information available from AHRIdirectory.org.

LEGEND

A-Weighted Sound Power Level (dBA)

UNIT SIZE	STANDARD RATING (dBA)	TYPICAL OCTAVE BAND SPECTRUM (dBA without tone adjustment)						
		125	250	500	1000	2000	4000	8000
24	68	54	58	61	64	60	55	53
36	74	58	64	69	70	66	60	53
48	71	59	61	64	68	62	58	53
60	77	60	64	71	74	64	60	54

NOTE: Tested in compliance with AHRI 270 but not listed with AHRI.

Physical Data

UNIT SIZE	24040	24060	36060	48090	60090
NOMINAL CAPACITY (ton)	2	2	3	4	5
SHIPPING WEIGHT lb	361	361	464	514	529
SHIPPING WEIGHT kg	163.6	163.6	210.2	233.1	239.8
COMPRESSORS	Scroll				
Quantity	1				
REFRIGERANT (R-454B)					
Quantity lb	5.5	5.5	8.0	10.25	11.25
Quantity (kg.)	2.49	2.49	3.63	4.65	5.1
MINIMUM CONDITIONED SPACE ARE (SQ FT)	91	91	122	167	182
REFRIGERANT METERING DEVICE	TXV				
OUTDOOR COIL					
Rows...Fins/in.	1...21	1...21	2...21	2...21	2...21
Face Area (sq ft)	13.6	13.6	13.6	19.4	21.4
OUTDOOR FAN					
Nominal Cfm	2500	2500	3000	3300	3600
Diameter in.	24	24	26	26	26
Diameter (mm)	609.6	609.6	600.4	660.4	660.4
Motor Hp (Rpm)	1/12 (810)	1/12 (810)	1/5 (810)	1/5 (810)	1/5 (810)
INDOOR COIL					
Rows...Fins/in.	3...17	3...17	3...17	3...17	3...17
Face Area (sq ft)	3.7	3.7	4.7	5.7	5.7
INDOOR BLOWER					
Required Minimum Dissipation Airflow (Cfm)	160	160	213	293	319
Nominal Low Stage Cooling Airflow (Cfm)	600	600	900	1200	1200
Nominal High Stage Cooling Airflow (Cfm)	800	800	1200	1600	1750
Size in. (D x W)	10x10	10x10	11x10	11x10	11x10
Size (mm) (D x W)	254x254	254x254	279.4x254	279.4x254	279.4x254
Motor HP (RPM)	1/2 (1050)	1/2 (1050)	3/4 (1000)	1.0 (1075)	1.0 (1075)
FURNACE SECTION*					
Burner Orifice No. (Qty...Drill Size)					
Natural Gas (Factory Installed)	2...44	3...44	3...44	3...38	3...38
Propane Gas	2...55	3...55	3...55	3...53	3...53
HIGH-PRESSURE SWITCH	650 +/- 15				
(psig) Cut-out Reset (Auto)	420 +/- 25				
LOSS-OF-CHARGE / LOW-PRESSURE SWITCH (psig)	50 +/- 7				
cut-out Reset (auto)	95 +/- 7				
RETURN-AIR FILTERS Throwaway [†] (filter inside home) in. (mm)	20x20x1 508x508x25	20x24x1 508x610x25	24x30x1 610x762x25	24x36x1 610x914x25	
RETURN-AIR FILTERS (Filter in accessory Internal filter Rack in unit) ^{†‡}					
Throwaway Size in. (mm)	2 each 12x20x1 2 each 305x508x25		1 each 14x24x1, 16x24x1 1 each 356x610x25, 406x610x25	1 each 16x24x1, 18x24x1 406x610x25, 457x610 x25	

*. Based on altitude of 0 to 2000 ft (0-610 m).

[†]. Required filter sizes shown are based on the larger of the AHRI (Air Conditioning Heating and Refrigeration Institute) rated cooling airflow or the heating airflow velocity of 300 ft/minute for throwaway type. Air filter pressure drop for non-standard filters must not exceed 0.08 IN. W.C.

[‡]. If using accessory filter rack refer to the filter rack installation instructions for correct filter sizes and quantity.

Accessories

Accessory Model Number	Description	Use With
CPRFCURB011B00*	Roof Curb, 14-in. High	24
CPRFCURB013B00	Roof Curb, 14-in. High	36 - 60
CPADCURB001A00†	Adapter curb	24
CPADCURB002A00†	Adapter curb	36 - 60
CPGSKTKIT001A00	Gasket Kit for existing roof curb with new base rail unit	24
CPMANDPR007A00	Manual Outside Air Damper - (Includes filter rack and 1-in. filter, same as CPFILTRK kit)	24
CPMANDPR008A00		36
CPMANDPR009A00		48 - 60
ECD-SDSML-JC2-ADB†	Vertical economizer with Jade Honeywell W7220 controller, Honeywell communicating actuator, and dry bulb sensor. (Contact MicroMetl Customer Service at 1-800-662-4822 to order)	24
ECD-SDLGS-JC2-ADB†		36
ECD-SDLGB-JC2-ADB†		48 - 60
ECH-SDSML-JC2-ADB†	Horizontal economizer with Jade Honeywell W7220 controller, Honeywell communicating actuator, and dry bulb sensor. (Contact MicroMetl Customer Service at 1-800-662-4822 to order)	24
ECH-SDLGS-JC2-ADB†		36
ECH-SDLGB-JC2-ADB†		48 - 60
CPFILTRK007A00	Internal Filter Rack (includes 1-in. filters)	24
CPFILTRK008A00		36
CPFILTRK009A00		48 - 60
KSALA0301410	Low Ambient Control (Pressure Switch)	All
NRTIMEGD001A00	Five Minute Compressor Delay	All
CPHSTART002A00	PTC Compressor Start Assist Kit	All
CPCRKHTR008A00	240V Crankcase Heater	24 - 36 Single Phase
CPCRKHTR004A00		48 - 60 Single Phase
Standard		36 - 60 3-Phase
CPFLUEDS004A00	Flue Discharge Deflector Assembly	All
DUCFLG002A00	Square to Round (1 set of 2, use with horizontal duct flanges only)	24 - 48

*. CPRFCURB011B00 can be used with 36-60 size units with some overhang.

†. Contact MicroMetal Customer Service at 1-800-662-4822 to order.

Unit Dimensions - 24

COOLING CAPACITY		UNIT WT.		UNIT HEIGHT IN/MM		CENTER OF GRAVITY IN/MM			
24		LB / KG		"A"		X		Y	
		350	158.6	45-3/4	1162	22-13/16	579.4	15-5/16	388.9
								15-13/16	401.6

COOLING CAPACITY		SHIPPING HEIGHT		SHIPPING LENGTH		SHIPPING WIDTH		SHIPPING WEIGHT	
24		IN / MM		IN / MM		IN / MM		LB / KG	
		51-1/2	1308	48	1219	32-1/2	826	361	163.6

UNIT		CORNER WEIGHT LB/KG			
		"1"	"2"	"3"	"4"
24		89.4	40.6	79.0	35.8
				85.6	38.8
				96.0	43.6

NOTE: 1. ALL TABLE DATA RELEVANT FOR ALL FACTORY INSTALLED OPTIONS EXCEPT ECONOMIZER.
2. * - INDICATES ALL TOP COILS FOR THE MODELS LISTED.

REQUIRED CLEARANCES TO COMBUSTIBLE MATL.

	INCHES (MM)
TOP OF UNIT	14 (355.6)
EVAP. COIL ACCESS SIDE	36 (914.0)
CONDENSER ACCESS SIDE	36 (914.0)
SIDE OPPOSITE DUCTS	48 (1219.2)
BOTTOM OF UNIT	172 (4371.2)
FLUE PANEL	36 (914.0)

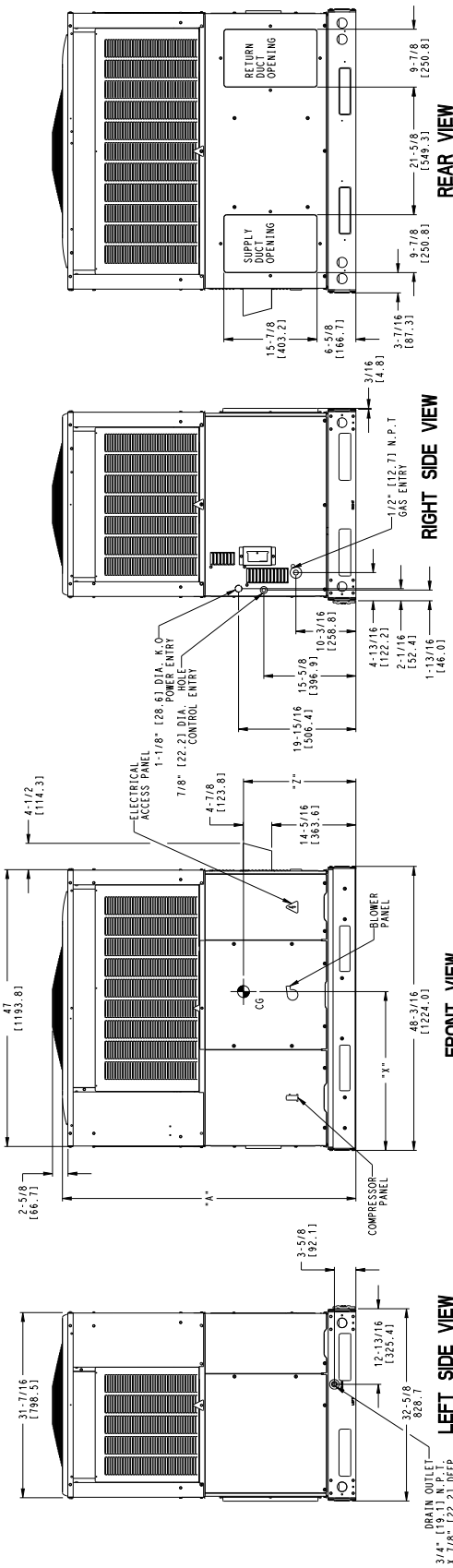
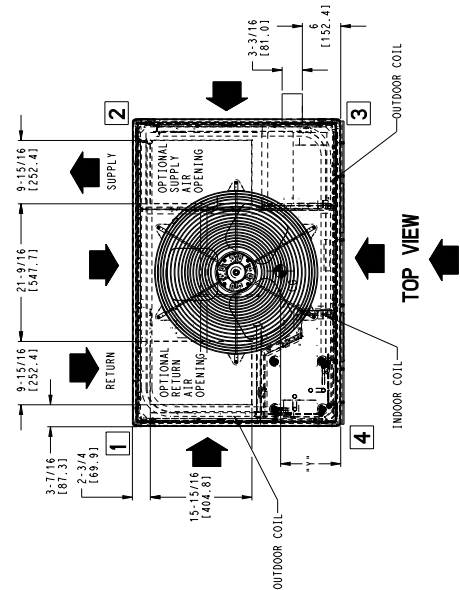
REQUIRED CLEARANCE FOR OPERATION AND SERVICE

	INCHES (MM)
EVAP. COIL ACCESS SIDE	36 (914.0)
CONDENSER ACCESS SIDE	36 (914.0)
SIDE OPPOSITE DUCTS	48 (1219.2)
UNIT TOP (EXCEPT FOR NEC REQUIREMENTS)	48 (1219.2)
SIDE OPPOSITE DUCTS	36 (914.0)
DUCT PANEL	12 (304.8)

NEC REQUIRED CLEARANCES

	INCHES (MM)
BETWEEN UNITS, POWER ENTRY SIDE	2 (50.8)
UNIT AND UNGROUND SURFACES, POWER ENTRY SIDE	38 (974.0)
UNIT AND BLOCK OR CONCRETE WALLS AND OTHER GROUND SURFACES, POWER ENTRY SIDE	42 (1066.8)

*MINIMUM DISTANCES: IF UNIT IS PLACED LESS THAN 12 (304.8) FROM WALL SYSTEM, THEN SYSTEM PERFORMANCE MAYBE COMPROMISED.
DIMENSIONS IN [] ARE IN MM



SD6111-4

Unit Dimensions - 36-60

COOLING CAPACITY	UNIT WT.		UNIT HEIGHT IN/MM		CENTER OF GRAVITY IN/MM			
	LB	KG	"A"		X	Y	Z	
36	450	203.9	44-3/4	1137	22-13/16	579.4	18	457.2 17-1/8 435.0
48	500	226.7	50-3/4	1289	22-13/16	579.4	18	457.2 17-3/8 441.3
60	515	233.5	52-3/4	1340	22-13/16	579.4	18	457.2 17-5/8 447.7

COOLING CAPACITY	SHIPPING HEIGHT		SHIPPING LENGTH		SHIPPING WIDTH		SHIPPING WEIGHT	
	IN	MM	IN	MM	IN	MM	LB	KG
36	48-3/4	1238	48-1/4	1226	44	1118	464	210.2
48	54-3/4	1391	48-1/4	1226	44	1118	514	233.1
60	56-3/4	1441	48-1/4	1226	44	1118	529	239.8

UNIT	CORNER WEIGHT LB/KG		
	"1"	"2"	"3"
36	98.2	44.5	87.0
48	108.7	49.3	96.8
60	111.5	50.6	99.2

NOTE: 1. ALL TABLE DATA RELEVANT FOR ALL FACTORY INSTALLED OPTIONS EXCEPT ECONOMIZER.
2. * - INDICATES ALL FTOP CODES FOR THE MODELS LISTED.

REQUIRED CLEARANCES TO COMBUSTIBLE MAIL

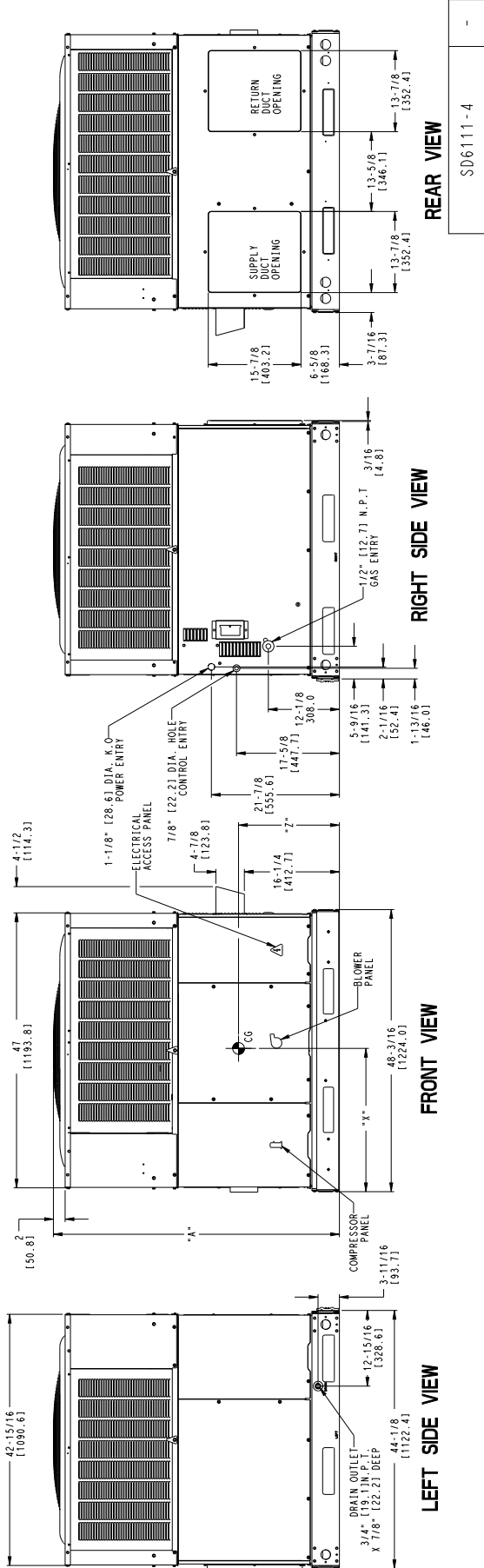
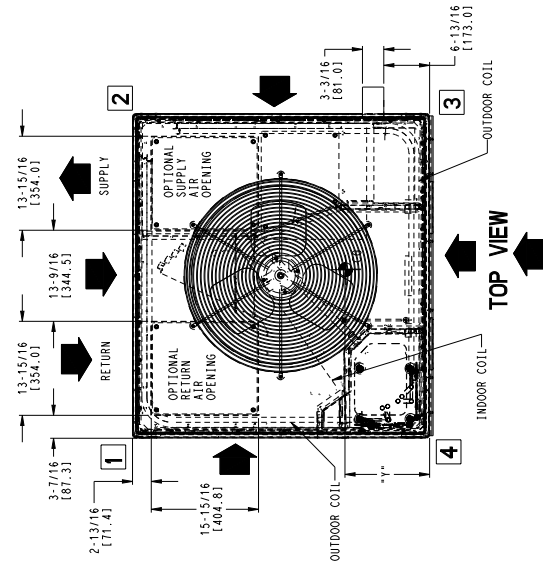
	INCHES [MM]
TOP OF UNIT.....	14 [355.6]
DUCT SIDE OF UNIT.....	2 [50.8]
DUCT OPPOSITE DUCTS.....	14 [355.6]
BOTTOM OF UNIT.....	172 [12.7]
FLUE PANEL.....	36 [914.4]

NEC REQUIRED CLEARANCES

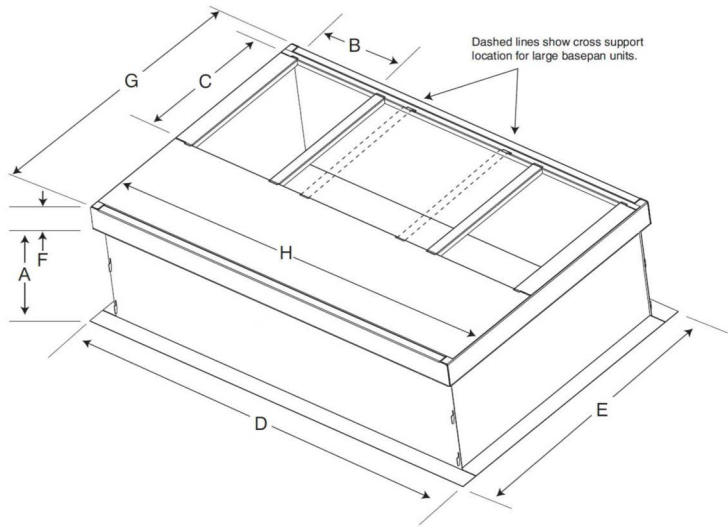
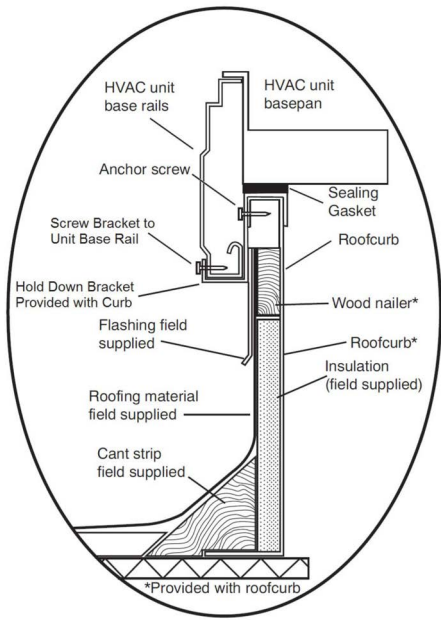
	INCHES (MM)
BETWEEN UNITS, POWER ENTRY SIDE.....	42 (1066.8)
UNIT AND UNGROUNDED SURFACES, POWER ENTRY SIDE.....	36 (914.0)
UNIT AND BLOCK OR CONCRETE WALLS AND OTHER	
GROUNDED SURFACES, POWER ENTRY SIDE.....	42 (1066.8)

REQUIRED CLEARANCE FOR OPERATION AND SERVICING		INCHES [MM]
EVAP. COIL ACCESS SIDE.....	36	[914.0]
POWER ENTRY SIDE.....	42	[1066.8]
(EXCEPT FOR NEC REQUIREMENTS)		
UNIT TOP.....	48	[1219.2]
SIDE OPPOSITE DUCTS.....	36	[914.0]
DUCT PANEL.....	12	[304.8]*

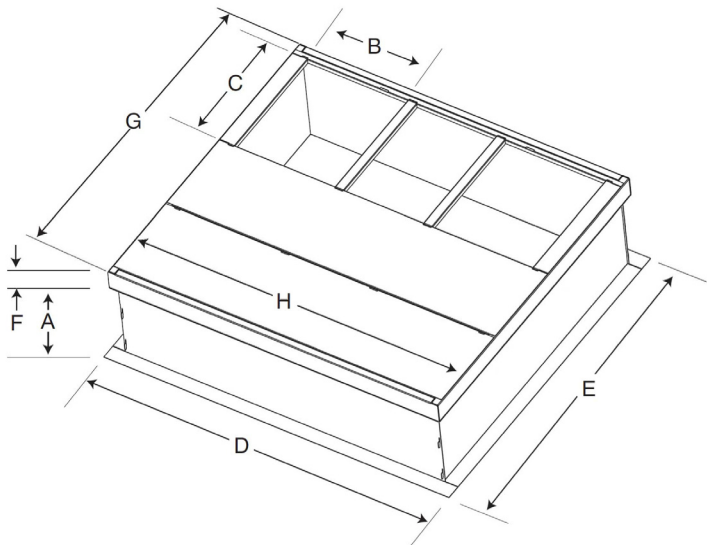
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DIMENSIONS IN () ARE IN MM



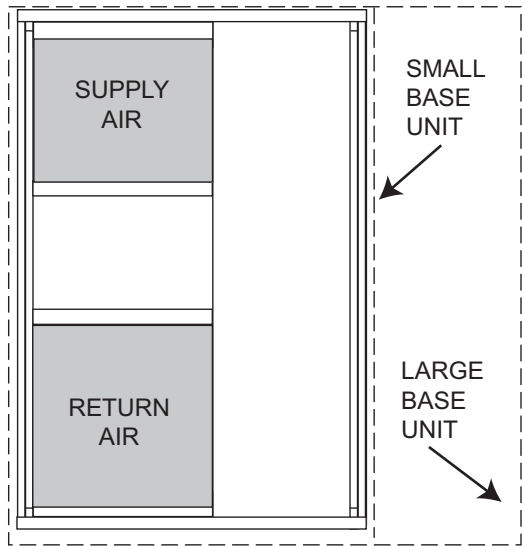
Accessory Dimensions



SMALL/COMMON CURB



LARGE CURB



UNIT PLACEMENT ON COMMON CURB

SMALL OR LARGE BASE UNIT

A180216

Unit Size	Catalog Number	A in. (mm)	B (small / common base) in. (mm)*	B (large base) in. (mm)*	C in. (mm)	D in. (mm)	E in. (mm)	F in. (mm)	G in. (mm)	H in. (mm)
Small or Large	CPRFCURB011B00	14 (356)	10 (254)	14 (356)	16 (406)	47.8 (1214)	32.4 (822)	2.7 (69)	30.6 (778)	46.1 (1170)
Large	CPRFCURB013B00	14 (356)	14 (356)				43.9 (1116)			

* Part Number CPRFCURB011B00 can be used on both small and large basepan units. The cross supports must be located based on whether the unit is a small basepan or a large basepan.

NOTES:

1. Roof curb must be set up for unit being installed.
2. Seal strip must be applied, as required, to unit being installed.
3. Roof curb is made of 16-gauge steel.
4. Attach ductwork to curb (flanges of duct rest on curb).
5. Insulated panels: 1-in. (25.4 mm) thick fiberglass 1 lb. density.

Selection Procedure (with example)

1. Determine cooling and heating requirements at design conditions:
Given:
Required Cooling Capacity (TC)34,000Btuh
Sensible Heat Capacity (SHC) 25,000 Btuh
Required Heating Capacity..... 45,000 Btuh
Condenser Entering Air Temperature 95°F (35°C)
Indoor-Air Temperature80°F (26°C)edb 67°F (19°C)ewb
Evaporator Air Quantity 1200 CFM
External Static Pressure 0.100 IN. W.C.
Electrical Characteristics208-1-60
2. Select unit based on required cooling capacity.
Enter Net Cooling Capacities table at condenser entering temperature of 95°F (35°C). Unit 036 at 1200 cfm and 67°F (19°C) ewb (entering wet bulb) will provide a total capacity of 34,800 Btuh and a SHC of 26,310 Btuh. Calculate SHC correction, if required, using Note 4 under Cooling Capacities tables.
3. Select heating capacity of unit to provide design condition requirement.
In the Heating Capacities and Efficiencies table, note that the unit 036060 (208/230 VAC) will provide 49,000 Btuh with an input of 60,000 Btuh.

4. Determine fan speed and power requirements at design conditions.
Before entering the air delivery tables, calculate the total static pressure required. From the given example, the Wet Coil Pressure Drop Table, and the Filter Pressure Drop Table:
- | | |
|--------------------------|---------------------|
| External Static Pressure | 0.100 IN. W.C |
| Filter | 0.06 IN. W.C |
| Wet Coil Pressure Drop | <u>0.06 IN. W.C</u> |
| Total Static Pressure | 0.22 IN. W.C |
- Enter the table for Dry Coil Air Delivery—Horizontal and Downflow Discharge. At .22 IN. W.C. ESP, the closest speed to 1200 CFM is Speed 6 (1223 CFM @ .4 in. W.C.ESP).
5. Select unit that corresponds to power source available.
The Electrical Data Table shows that the unit is designed to operate at 208-1-60.

Performance Data
024 Low Cool

EVAPORATOR AIR		CONDENSER ENTERING AIR TEMPERATURES °F (°C)																	
CFM / BF	EWB	75 (23.9)			85 (29.4)			95 (35)			105 (40.6)			115 (46.1)			125 (51.7)		
		Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW
		Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens	
525 / .04	57 (13.8)	17.0	17.0	1.0	16.0	16.0	1.1	14.7	14.7	1.3	13.2	13.2	1.4	11.7	11.7	1.7	10.8	10.8	1.8
	62 (16.6)	17.6	15.8	1.0	16.4	15.2	1.1	15.1	15.1	1.3	13.6	13.6	1.4	12.0	12.0	1.7	11.1	11.1	1.8
	63* (17.2)	17.9	12.8	1.0	16.8	12.4	1.1	15.2	11.6	1.3	13.5	10.8	1.4	11.8	10.1	1.7	10.7	9.5	1.8
	67 (19.4)	19.3	13.3	1.0	18.0	12.9	1.1	16.4	12.1	1.3	14.6	11.3	1.4	12.7	10.5	1.7	11.5	9.9	1.8
	72 (22.2)	21.0	10.6	1.0	19.6	10.2	1.1	17.9	9.5	1.3	15.9	8.8	1.5	13.9	8.1	1.7	12.5	7.4	1.9
600 / .05	57 (13.8)	16.7	16.7	1.0	15.8	15.8	1.1	14.5	14.5	1.3	13.0	13.0	1.5	11.5	11.5	1.7	10.6	10.5	1.9
	62 (16.6)	16.9	16.5	1.0	15.8	15.8	1.1	14.5	14.5	1.3	13.0	13.0	1.5	11.5	11.5	1.7	10.6	10.5	1.9
	63* (17.2)	17.1	13.3	1.0	16.0	12.8	1.1	14.5	12.0	1.3	12.9	11.3	1.5	11.3	10.5	1.7	10.2	9.8	1.9
	67 (19.4)	18.4	13.8	1.0	17.2	13.4	1.1	15.6	12.5	1.3	13.9	11.7	1.5	12.1	10.9	1.7	11.0	10.2	1.9
675 / .06	72 (22.2)	20.0	11.1	1.0	18.7	10.6	1.1	17.0	9.9	1.3	15.1	9.2	1.5	13.2	8.4	1.7	11.9	7.7	1.9
	57 (13.8)	18.2	17.5	1.0	17.2	16.6	1.1	15.8	15.5	1.3	14.2	13.9	1.5	12.5	12.3	1.7	11.6	11.4	1.9
	62 (16.6)	18.4	17.2	1.0	17.2	16.6	1.1	15.8	15.6	1.3	14.2	13.9	1.5	12.5	12.3	1.7	11.6	11.4	1.9
	63* (17.2)	18.6	13.9	1.0	17.4	13.5	1.1	15.8	12.9	1.3	14.1	12.0	1.5	12.3	11.2	1.7	11.1	10.6	1.9
	67 (19.4)	20.0	14.5	1.0	18.7	14.1	1.2	17.0	13.5	1.3	15.1	12.5	1.5	13.2	11.2	1.7	12.0	11.1	1.9
720 / .06	72 (22.2)	21.9	11.6	1.0	20.4	11.1	1.2	18.6	10.6	1.3	16.5	9.8	1.5	14.4	9.0	1.7	13.0	8.3	1.9

See Legend and Notes on page 15.

024 High Cool

EVAPORATOR AIR		CONDENSER ENTERING AIR TEMPERATURES °F (°C)																	
CFM / BF	EWB °F (°C)	75 (23.9)			85 (29.4)			95 (35)			105 (40.6)			115 (46.1)			125 (51.7)		
		Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW
		Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens	
700 / 0.08	57 (13.8)	23.4	23.4	1.6	22.4	22.4	1.8	20.6	20.6	1.9	19.2	19.2	2.1	17.2	17.2	2.4	16.3	16.3	2.7
	62 (16.6)	24.3	22.4	1.6	23.0	21.8	1.8	21.1	20.1	1.9	19.8	18.9	2.1	17.7	16.5	2.4	16.8	14.8	2.7
	63* (17.2)	24.7	18.1	1.6	23.5	17.4	1.8	21.4	16.4	1.9	19.7	16.0	2.1	17.5	14.6	2.4	16.2	13.8	2.7
	67 (19.4)	26.6	18.8	1.6	25.2	18.2	1.8	22.9	17.1	1.9	21.2	16.7	2.1	18.8	15.2	2.4	17.4	14.4	2.7
	72 (22.2)	29.0	15.1	1.6	27.5	14.3	1.8	25.1	13.5	2.0	23.1	13.0	2.2	20.5	11.7	2.4	19.0	10.9	2.8
800 / 0.09	57 (13.8)	24.5	24.3	1.6	23.5	22.8	1.8	21.6	20.9	2.0	20.2	19.6	2.2	18.1	17.2	2.4	17.1	15.4	2.8
	62 (16.6)	24.8	23.3	1.6	23.5	22.6	1.8	21.6	20.9	2.0	20.2	19.6	2.2	18.1	17.2	2.4	17.1	15.4	2.8
	63* (17.2)	25.0	18.8	1.6	23.7	18.1	1.8	21.6	17.1	2.0	20.0	16.7	2.2	17.7	15.1	2.4	16.4	14.4	2.8
	67 (19.4)	26.9	19.6	1.6	25.5	18.9	1.8	23.2	17.8	2.0	21.5	17.4	2.2	19.0	15.8	2.4	17.6	15.0	2.8
	72 (22.2)	29.3	15.7	1.6	27.8	14.9	1.8	25.4	14.0	2.0	23.4	13.5	2.2	20.7	12.1	2.4	19.2	11.4	2.8
900 / 0.09	57 (13.8)	26.4	25.9	1.6	25.4	24.4	1.8	23.3	22.4	2.0	21.8	21.0	2.2	19.5	18.4	2.4	18.5	16.5	2.8
	62 (16.6)	26.7	25.2	1.6	25.4	24.2	1.8	23.3	22.4	2.0	21.8	21.0	2.2	19.5	18.4	2.4	18.5	16.5	2.8
	63* (17.2)	27.0	20.1	1.6	25.6	19.4	1.8	23.3	18.3	2.0	21.6	17.8	2.2	19.1	16.2	2.4	17.7	15.3	2.8
	67 (19.4)	29.1	20.9	1.7	27.6	20.2	1.8	25.1	19.0	2.0	23.2	18.5	2.2	20.5	16.8	2.4	19.0	16.0	2.8
960 / 0.09	72 (22.2)	31.7	16.7	1.7	30.0	15.9	1.9	27.4	15.0	2.0	25.3	14.5	2.2	22.4	13.0	2.5	20.8	12.1	2.8

See Legend and Notes on page 15.

Manufacturer reserves the right to change, at any time, specifications and designs without notice and without obligations.

Performance Data (cont)
036 Low Cool

EVAPORATOR AIR		CONDENSER ENTERING AIR TEMPERATURES °F (°C)																	
		75 (23.9)			85 (29.4)			95 (35)			105 (40.6)			115 (46.1)			125 (51.7)		
		Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW
CFM / BF	EWB	Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens	
775 / .04	57 (13.8)	27.4	27.4	1.5	25.7	25.7	1.7	24.0	24.0	1.9	22.2	22.2	2.2	20.2	20.2	2.5	18.2	18.2	2.8
	62 (16.6)	28.4	24.6	1.5	26.4	23.6	1.7	24.7	24.7	1.9	22.8	22.8	2.2	20.7	20.7	2.5	18.7	18.7	2.8
	63* (17.2)	29.0	19.9	1.5	26.9	18.9	1.7	24.9	18.0	1.9	22.7	16.9	2.2	20.5	15.9	2.5	18.1	14.9	2.8
	67 (19.4)	31.1	20.7	1.5	29.0	19.7	1.7	26.8	18.8	1.9	24.4	17.6	2.2	22.0	16.6	2.5	19.4	15.5	2.8
	72 (22.2)	34.0	16.6	1.5	31.6	15.6	1.7	29.3	14.8	2.0	26.6	13.8	2.2	24.0	12.7	2.5	21.2	11.7	2.8
900 / .05	57 (13.8)	27.4	27.1	1.5	25.8	25.2	1.7	24.1	23.3	2.0	22.2	21.1	2.2	20.2	19.0	2.5	18.2	16.9	2.9
	62 (16.6)	27.7	26.0	1.5	25.8	25.0	1.7	24.1	23.3	2.0	22.2	21.1	2.2	20.2	19.0	2.5	18.2	16.9	2.9
	63* (17.2)	28.0	21.0	1.5	26.1	20.0	1.7	24.1	19.0	2.0	22.0	17.9	2.2	19.8	16.8	2.5	17.5	15.8	2.9
	67 (19.4)	30.1	21.9	1.5	28.0	20.8	1.7	25.9	19.8	2.0	23.6	18.6	2.2	21.3	17.5	2.5	18.8	16.4	2.9
	72 (22.2)	32.9	17.5	1.6	30.5	16.4	1.8	28.3	15.6	2.0	25.8	14.5	2.3	23.2	13.5	2.6	20.5	12.3	2.9
1025 / .06	57 (13.8)	30.3	28.7	1.6	28.5	26.8	1.8	26.6	25.3	2.0	24.6	22.8	2.3	22.3	20.6	2.6	20.2	18.5	2.9
	62 (16.6)	30.6	27.6	1.6	28.5	26.6	1.8	26.6	25.3	2.0	24.6	22.8	2.3	22.3	20.6	2.6	20.2	18.5	2.9
	63* (17.2)	31.0	22.2	1.6	28.8	21.3	1.8	26.6	20.7	2.0	24.3	19.4	2.3	21.9	18.1	2.6	19.3	17.2	2.9
	67 (19.4)	33.3	23.2	1.6	31.0	22.1	1.8	28.6	21.5	2.0	26.1	20.2	2.3	23.5	18.1	2.6	20.8	17.9	2.9
	72 (22.2)	36.3	18.5	1.6	33.7	17.5	1.8	31.3	17.0	2.0	28.5	15.7	2.3	25.6	14.6	2.6	22.7	13.5	2.9

See Legend and Notes on page 15.

036 High Cool

EVAPORATOR AIR		CONDENSER ENTERING AIR TEMPERATURES °F (°C)																	
		75 (23.9)			85 (29.4)			95 (35)			105 (40.6)			115 (46.1)			125 (51.7)		
		Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW
CFM / BF	EWB °F (°C)	Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens	
1050 / 0.08	57 (13.8)	34.7	34.7	2.4	32.3	32.3	2.5	30.3	30.3	2.8	29.7	29.7	3.1	27.4	27.4	3.5	25.0	25.0	3.9
	62 (16.6)	36.1	33.0	2.4	33.1	31.6	2.5	31.2	29.6	2.8	30.6	27.4	3.1	28.2	25.1	3.5	25.7	22.0	3.9
	63* (17.2)	36.8	26.6	2.4	33.8	25.3	2.5	31.5	24.2	2.8	30.5	23.2	3.1	27.8	22.1	3.5	24.8	20.5	3.9
	67 (19.4)	39.6	27.7	2.4	36.4	26.4	2.6	33.8	25.2	2.9	32.8	24.2	3.1	29.9	23.0	3.5	26.7	21.4	3.9
	72 (22.2)	43.1	22.2	2.5	39.6	20.8	2.6	37.0	19.8	2.9	35.8	18.9	3.1	32.6	17.7	3.5	29.1	16.2	3.9
1200 / 0.09	57 (13.8)	36.4	35.7	2.5	33.8	33.1	2.6	31.8	30.8	2.9	31.2	28.4	3.1	28.8	26.0	3.5	26.2	22.9	3.9
	62 (16.6)	36.8	34.3	2.5	33.8	32.9	2.6	31.8	30.8	2.9	31.2	28.4	3.1	28.8	26.0	3.5	26.2	22.9	3.9
	63* (17.2)	37.2	27.7	2.5	34.2	26.3	2.6	31.8	25.2	2.9	30.9	24.1	3.1	28.1	22.9	3.5	25.1	21.3	3.9
	67 (19.4)	40.0	28.8	2.5	36.8	27.4	2.6	34.2	26.2	2.9	33.2	25.2	3.1	30.3	23.9	3.5	27.0	22.2	3.9
	72 (22.2)	43.6	23.1	2.5	40.1	21.6	2.6	37.4	20.6	2.9	36.2	19.6	3.1	33.0	18.4	3.6	29.4	16.9	4.0
1350 / 0.10	57 (13.8)	37.7	37.7	2.5	35.0	35.0	2.6	32.9	32.9	2.9	32.3	30.4	3.2	29.8	27.8	3.6	27.1	24.4	4.0
	62 (16.6)	38.1	37.0	2.5	35.0	35.0	2.6	32.9	32.9	2.9	32.3	30.4	3.2	29.8	27.8	3.6	27.1	24.4	4.0
	63* (17.2)	38.5	29.6	2.5	35.4	28.1	2.6	33.0	26.9	2.9	31.9	25.8	3.2	29.1	24.5	3.6	26.0	22.8	4.0
	67 (19.4)	41.4	30.8	2.5	38.1	29.3	2.7	35.4	28.0	2.9	34.3	26.9	3.2	31.3	25.5	3.6	28.0	23.7	4.0
	72 (22.2)	45.1	24.6	2.6	41.5	23.1	2.7	38.7	22.0	3.0	37.4	21.0	3.2	34.1	19.7	3.6	30.5	18.0	4.0

See Legend and Notes on page 15.

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Performance Data (cont)
048 Low Cool

EVAPORATOR AIR		CONDENSER ENTERING AIR TEMPERATURES °F (°C)																	
		75 (23.9)			85 (29.4)			95 (35)			105 (40.6)			115 (46.1)			125 (51.7)		
		Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total V KW	Capacity MBtuh		Total Syst KW
CFM / BF	EWB	Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens	
1050 / .05	57 (13.8)	36.9	36.9	1.8	35.0	35.0	2.1	32.6	32.6	2.4	29.9	29.9	2.8	27.2	27.2	3.2	24.0	24.0	3.5
	62 (16.6)	38.3	34.0	1.8	35.9	32.9	2.1	33.5	33.5	2.4	30.7	30.7	2.8	27.9	27.9	3.2	24.6	24.6	3.5
	63* (17.2)	39.0	27.5	1.8	36.6	26.3	2.1	33.8	25.3	2.4	30.6	24.0	2.8	27.5	22.8	3.2	23.8	21.3	3.5
	67 (19.4)	42.0	28.6	1.8	39.4	27.4	2.1	36.3	26.3	2.5	32.9	25.0	2.8	29.6	23.7	3.2	25.6	22.2	3.5
1200 / .06	72 (22.2)	45.8	22.9	1.9	42.9	21.6	2.1	39.7	20.7	2.5	36.0	19.5	2.8	32.3	18.3	3.2	27.9	16.6	3.6
	57 (13.8)	36.4	36.4	1.9	34.5	34.4	2.1	32.1	32.1	2.5	29.5	29.4	2.8	26.8	26.8	3.2	23.6	23.6	3.6
	62 (16.6)	36.8	35.4	1.9	34.5	34.2	2.1	32.1	32.1	2.5	29.5	29.4	2.8	26.8	26.8	3.2	23.6	23.6	3.6
	63* (17.2)	37.2	28.5	1.9	34.9	27.3	2.1	32.2	26.3	2.5	29.2	24.9	2.8	26.2	23.7	3.2	22.7	22.1	3.6
1350 / .06	67 (19.4)	40.0	29.7	1.9	37.5	28.5	2.2	34.6	27.4	2.5	31.4	26.0	2.8	28.2	24.7	3.2	24.4	23.0	3.6
	72 (22.2)	43.6	23.8	1.9	40.8	22.5	2.2	37.8	21.5	2.5	34.2	20.3	2.9	30.7	19.0	3.2	26.5	17.3	3.6
	57 (13.8)	39.7	38.1	1.9	37.6	36.2	2.2	35.0	34.5	2.5	32.2	31.4	2.9	29.2	28.6	3.2	25.8	25.5	3.6
	62 (16.6)	40.1	37.0	1.9	37.6	35.9	2.2	35.1	34.5	2.5	32.2	31.4	2.9	29.2	28.6	3.2	25.8	25.5	3.6
	63* (17.2)	40.6	29.8	1.9	38.1	28.7	2.2	35.1	28.2	2.5	31.8	26.7	2.9	28.6	25.3	3.2	24.7	23.9	3.6
	67 (19.4)	43.6	31.1	1.9	40.9	29.9	2.2	37.7	29.4	2.6	34.2	27.8	2.9	30.8	25.2	3.2	26.6	24.9	3.6
	72 (22.2)	47.6	24.9	1.9	44.6	23.7	2.2	41.2	23.1	2.6	37.4	21.7	2.9	33.6	20.3	3.3	29.0	18.7	3.7

See Legend and Notes on page 15.

048 High Cool

EVAPORATOR AIR		CONDENSER ENTERING AIR TEMPERATURES °F (°C)																	
		75 (23.9)			85 (29.4)			95 (35)			105 (40.6)			115 (46.1)			125 (51.7)		
		Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW
CFM / BF	EWB °F (°C)	Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens	
1400 / 0.09	57 (13.8)	42.4	42.4	3.0	42.5	42.5	3.4	41.6	41.6	3.7	39.6	39.6	4.0	37.0	37.0	4.6	34.4	34.4	5.0
	62 (16.6)	44.1	43.3	3.0	43.6	42.9	3.4	42.8	41.2	3.7	40.7	38.4	4.0	38.1	35.5	4.6	35.3	32.5	5.0
	63* (17.2)	44.9	35.7	3.0	44.5	35.3	3.4	43.3	33.6	3.7	40.6	32.6	4.0	37.6	31.3	4.6	34.1	30.3	5.0
	67 (19.4)	48.3	37.1	3.1	47.9	36.8	3.4	46.5	35.0	3.7	43.7	34.0	4.0	40.4	32.6	4.6	36.7	31.5	5.0
	72 (22.2)	52.7	29.7	3.1	52.2	29.1	3.4	50.8	27.6	3.8	47.6	26.5	4.0	44.1	25.1	4.6	40.0	24.0	5.0
1600 / 0.10	57 (13.8)	44.5	44.5	3.1	44.5	44.5	3.4	43.7	42.8	3.8	41.5	39.9	4.1	38.8	36.9	4.6	36.0	33.7	5.1
	62 (16.6)	45.0	45.0	3.1	44.5	44.5	3.4	43.7	42.8	3.8	41.5	39.9	4.1	38.8	36.9	4.6	36.0	33.7	5.1
	63* (17.2)	45.5	37.0	3.1	45.0	36.7	3.4	43.8	34.9	3.8	41.1	33.9	4.1	38.0	32.5	4.6	34.5	31.4	5.1
	67 (19.4)	48.9	38.6	3.1	48.4	38.2	3.5	47.0	36.4	3.8	44.2	35.3	4.1	40.9	33.9	4.6	37.1	32.8	5.1
	72 (22.2)	53.3	30.9	3.2	52.8	30.2	3.5	51.4	28.6	3.8	48.2	27.5	4.1	44.6	26.1	4.7	40.5	24.9	5.1
1800 / 0.11	57 (13.8)	46.0	46.0	3.2	46.1	46.1	3.5	45.2	45.2	3.8	43.0	42.6	4.1	40.2	39.4	4.7	37.3	36.1	5.1
	62 (16.6)	46.5	46.5	3.2	46.1	46.1	3.5	45.2	45.2	3.8	43.0	42.6	4.1	40.2	39.4	4.7	37.3	36.1	5.1
	63* (17.2)	47.0	39.6	3.2	46.6	39.2	3.5	45.3	37.3	3.8	42.5	36.2	4.1	39.4	34.7	4.7	35.7	33.6	5.1
	67 (19.4)	50.6	41.2	3.2	50.1	40.8	3.5	48.6	38.9	3.9	45.7	37.7	4.1	42.3	36.2	4.7	38.4	35.0	5.1
	72 (22.2)	55.1	33.0	3.2	54.6	32.3	3.6	53.2	30.6	3.9	49.8	29.4	4.2	46.1	27.9	4.7	41.9	26.6	5.2

See Legend and Notes on page 15.

Manufacturer reserves the right to change, at any time, specifications and designs without notice and without obligations.

Performance Data (cont)
060 Low Cool

EVAPORATOR AIR		CONDENSER ENTERING AIR TEMPERATURES °F (°C)																	
		75 (23.9)			85 (29.4)			95 (35)			105 (40.6)			115 (46.1)			125 (51.7)		
		Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW	Capacity MBtuh		Total Syst KW
CFM / BF	EWB	Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens	
1050 / .05	57 (13.8)	39.2	39.2	2.3	36.9	36.9	2.7	34.3	34.3	3.0	31.3	31.3	3.5	28.2	28.2	4.0	25.3	25.3	4.4
	62 (16.6)	40.8	36.3	2.3	37.9	35.1	2.7	35.3	35.3	3.0	32.2	32.2	3.5	29.0	29.0	4.0	26.0	26.0	4.4
	63* (17.2)	41.6	29.3	2.3	38.7	28.1	2.7	35.7	26.5	3.0	32.2	24.8	3.5	28.7	23.2	4.0	25.2	21.8	4.4
	67 (19.4)	44.7	30.5	2.3	41.6	29.3	2.7	38.3	27.6	3.0	34.6	25.9	3.5	30.8	24.2	4.0	27.1	22.7	4.4
1200 / .06	72 (22.2)	48.7	24.4	2.3	45.3	23.1	2.8	41.9	21.7	3.1	37.7	20.2	3.6	33.6	18.6	4.0	29.5	17.0	4.4
	57 (13.8)	41.1	39.3	2.3	38.7	36.8	2.7	36.0	33.7	3.1	32.9	30.4	3.6	29.6	27.4	4.0	26.6	24.3	4.4
	62 (16.6)	41.6	37.7	2.3	38.7	36.5	2.7	36.0	33.7	3.1	32.9	30.4	3.6	29.6	27.4	4.0	26.6	24.3	4.4
	63* (17.2)	42.0	30.4	2.3	39.1	29.2	2.7	36.1	27.5	3.1	32.5	25.8	3.6	29.0	24.1	4.0	25.5	22.6	4.4
1350 / .06	67 (19.4)	45.2	31.7	2.3	42.1	30.4	2.8	38.8	28.7	3.1	35.0	26.9	3.6	31.2	25.1	4.0	27.4	23.6	4.4
	72 (22.2)	49.3	25.4	2.4	45.9	24.0	2.8	42.4	22.6	3.1	38.1	21.0	3.6	34.0	19.3	4.1	29.8	17.7	4.5
	57 (13.8)	42.6	42.0	2.4	40.1	39.3	2.8	37.3	36.0	3.1	34.0	32.5	3.6	30.7	29.3	4.1	27.5	25.9	4.5
	62 (16.6)	43.1	40.7	2.4	40.1	39.0	2.8	37.3	36.0	3.1	34.0	32.5	3.6	30.7	29.3	4.1	27.5	25.9	4.5
	63* (17.2)	43.5	32.5	2.4	40.5	31.2	2.8	37.3	29.4	3.1	33.7	27.6	3.6	30.0	25.8	4.1	26.4	24.2	4.5
	67 (19.4)	46.8	33.9	2.4	43.5	32.5	2.8	40.1	30.6	3.1	36.2	28.7	3.6	32.3	26.8	4.1	28.3	25.2	4.5
	72 (22.2)	51.0	27.1	2.4	47.5	25.7	2.8	43.8	24.1	3.2	39.4	22.4	3.7	35.2	20.7	4.1	30.9	18.9	4.5

See Legend and Notes on page 15.

060 High Cool

EVAPORATOR AIR		CONDENSER ENTERING AIR TEMPERATURES °F (°C)																	
		75 (23.9)			85 (29.4)			95 (35)			105 (40.6)			115 (46.1)			125 (51.7)		
		Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW	Capacity MBtuh		Total Sys KW
CFM / BF	EWB °F (°C)	Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens		Total	Sens	
1600 / 0.10	57 (13.8)	54.7	54.7	3.9	52.3	52.3	4.3	49.0	49.0	4.7	46.0	46.0	5.2	42.2	42.2	5.8	38.3	38.3	6.2
	62 (16.6)	56.9	50.7	3.9	53.8	48.9	4.3	50.5	45.9	4.7	47.3	41.9	5.2	43.4	38.8	5.8	39.4	34.2	6.2
	63* (17.2)	58.1	40.9	3.9	54.9	39.1	4.3	51.0	37.5	4.7	47.3	35.6	5.2	42.9	34.1	5.8	38.1	31.9	6.2
	67 (19.4)	62.5	42.6	3.9	59.1	40.8	4.3	54.8	39.1	4.8	50.8	37.1	5.2	46.1	35.6	5.8	41.0	33.2	6.2
	72 (22.2)	68.1	34.1	3.9	63.2	32.2	4.3	56.4	30.7	4.8	55.4	29.0	5.3	50.3	27.4	5.9	44.1	25.2	6.3
1750 / 0.10	57 (13.8)	58.1	55.0	3.9	55.5	51.4	4.3	52.0	47.8	4.8	48.8	43.7	5.3	44.8	40.4	5.9	40.6	35.6	6.3
	62 (16.6)	58.7	52.8	3.9	55.5	50.9	4.3	52.1	47.8	4.8	48.8	43.7	5.3	44.8	40.4	5.9	40.6	35.6	6.3
	63* (17.2)	59.4	42.6	3.9	56.1	40.8	4.3	52.1	39.1	4.8	48.3	37.1	5.3	43.9	35.6	5.9	39.0	33.2	6.3
	67 (19.4)	63.8	44.3	4.0	60.4	42.5	4.3	56.0	40.7	4.8	52.0	38.7	5.3	47.2	37.0	5.9	41.9	34.6	6.3
	72 (22.2)	69.6	35.5	4.0	65.8	33.5	4.4	61.2	32.0	4.8	56.6	30.2	5.3	51.4	28.5	5.9	45.0	26.3	6.4
2000 / 0.13	57 (13.8)	59.8	59.7	4.0	57.1	55.8	4.4	53.5	51.9	4.8	50.2	47.4	5.4	46.1	43.8	5.9	41.8	38.7	6.4
	62 (16.6)	60.4	57.9	4.0	57.1	55.3	4.4	53.6	51.9	4.8	50.2	47.4	5.4	46.1	43.8	5.9	41.8	38.7	6.4
	63* (17.2)	61.1	46.2	4.0	57.7	44.2	4.4	53.6	42.4	4.8	49.7	40.3	5.4	45.1	38.6	5.9	40.1	36.1	6.4
	67 (19.4)	65.7	48.1	4.0	62.1	46.1	4.4	57.6	44.2	4.9	53.4	42.0	5.4	48.5	40.2	5.9	43.1	37.6	6.4
	72 (22.2)	71.6	38.5	4.1	67.7	36.4	4.5	63.0	34.8	4.9	58.3	32.7	5.4	52.9	31.0	6.0	46.3	28.5	6.4

See Legend and Notes on page 15.

Performance Data (Cont)

*At 75°F (24°C) entering dry bulb-Tennessee Valley Authority (TVA) rating conditions; all others at 80°F (27°C) dry bulb.

LEGEND
BF— Bypass Factor
edb— Entering Dry-Bulb
ewb— Entering Wet-Bulb
KW — Total Unit Power Input
rh—Relative Humidity

COOLING NOTES:
1. Ratings are net; they account for the effects of the evaporator-fan motor power and heat.
2. Direct interpolation is permissible. Do not extrapolate.
3. The following formulas may be used:

$$t_{ldb} = t_{edb} - \frac{\text{Sensible Capacity (Btuh)}}{1.09 \times \text{CFM}}$$
$$t_{lwb} = \text{Wet-bulb temperature corresponding to enthalpy air leaving evaporator coil (h}_{lwb}\text{)}$$
$$h_{lwb} = h_{ewb} - \frac{\text{Total Capacity (Btuh)}}{(4.5 \times \text{CFM})}$$

Where: h_{ewb} = Enthalpy of air entering evaporator coil

4. The sensible capacities is based on 80° F (26.6°C) edb temperature of air entering evaporator coil. Below 80°F (26.6°C) edb, subtract (corr factor x cfm) from sensible capacities. Above 80°F (26.6°C) edb, add (corr factor x cfm) to sensible capacities.
Correction Factor = 1.09 x (1 - BF) x (t_{edb} - 80)

Gas Adjustment
Natural Gas Orifice Sizes and Manifold Pressure
Single and Three Phase Models

Nameplate Input (Btu/hr)		Altitude of Installation (FT. Above Sea Level) U.S.A.*
		0 to 2000 (0-610 m)
45,000	Orifice No. (Qty)	0.125-in.
	Manifold Press. (in. W.C.)	3.2~3.8
60,000	Orifice No. (Qty)	28 (1)
	Manifold Press. (in. W.C.)	3.2~3.8
89,000	Orifice No. (Qty)	18 (1)
	Manifold Press. (in. W.C.)	3.2~3.8

*. Altitude of 2001ft and above is not allowed.

High Altitude Compensation, Natural Gas - Single and Three Phase Models

Nameplate Input (Btu/hr)	Rated Heating Input (Btu/hr), Natural Gas at Installation Altitude Above Sea Level, U.S.A.*
	0 to 2000 ft (0-610 m)
45,000	45,000
60,000	60,000
89,000	89,000

*. Altitude of 2001 ft and above is not allowed.

Dry Coil Air Delivery** - Horizontal and Downflow Discharge Sizes 24-60 208/230 VAC - 1,3 Phase

Unit Size	Heating Rise °F (°C)	Motor Speed	Allowable Functions	Motor Speed Selection		ESP (in. W.C.)										
							0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
24040	25 - 55 (14 - 31)	1	Continuous Fan*	SW2-5	SW2-6	CFM	712	625	531	440	344	208	NA	NA	NA	NA
				OFF	OFF											
			Dehumidification Low	SW1-7	SW1-8											
				OFF	OFF	BHP	0.09	0.10	0.10	0.10	0.11	0.11	NA	NA	NA	NA
			Low Stage Cooling	SW1-3	SW1-4											
				OFF	OFF											
		2	Continuous Fan	SW2-5	SW2-6	CFM	512	460	344	212	NA	NA	NA	NA	NA	NA
				ON	OFF											
			Dehumidification Low	SW1-7	SW1-8											
				ON	OFF	BHP	0.06	0.06	0.07	0.07	NA	NA	NA	NA	NA	NA
			Low Stage Cooling	SW1-3	SW1-4											
				ON	OFF											
		3	Dehumidification Low	SW1-7	SW1-8	CFM	888	818	750	679	606	511	425	320	238	NA
				OFF	ON											
			Low Stage Cooling	SW1-3	SW1-4	BHP	0.18	0.18	0.19	0.19	0.20	0.20	0.21	0.21	0.22	NA
				OFF	ON											
		4	Dehumidification Low	SW1-7	SW1-8	CFM	768	686	604	520	430	327	210	115	NA	NA
				ON	ON											
			Low Stage Cooling*	SW1-3	SW1-4	BHP	0.11	0.11	0.12	0.12	0.13	0.13	0.13	0.13	NA	NA
				ON	ON											
24040	25 - 55 (14 - 31)	5	Dehumidification High	SW1-5	SW1-6	CFM	900	824	753	680	604	528	446	344	240	130
				OFF	OFF											
			High Stage Cooling	SW1-1	SW1-2	BHP	0.15	0.15	0.16	0.17	0.17	0.18	0.18	0.19	0.19	0.19
				OFF	OFF											
			Heating*	SW2-1	SW2-2	Gas Heat Rise (°F)	37	41	44	49	55	NA	NA	NA	NA	NA
				OFF	OFF	Gas Heat Rise (°C)	21	23	25	27	31	NA	NA	NA	NA	NA
		6	Dehumidification High	SW1-5	SW1-6	CFM	1063	994	929	866	803	741	673	610	528	453
				ON	OFF											
			High Stage Cooling	SW1-1	SW1-2	BHP	0.21	0.22	0.23	0.23	0.24	0.25	0.25	0.26	0.26	0.27
				ON	OFF											
			Heating	SW2-1	SW2-2	Gas Heat Rise (°F)	31	34	36	39	42	45	50	55	NA	NA
				ON	OFF	Gas Heat Rise (°C)	17	19	20	21	23	25	28	30	NA	NA
		7	Dehumidification High	SW1-5	SW1-6	CFM	956	883	817	747	676	604	529	450	348	241
				OFF	ON											
			High Stage Cooling	SW1-1	SW1-2	BHP	0.17	0.18	0.18	0.19	0.19	0.20	0.20	0.21	0.21	0.22
				OFF	ON											
				SW2-1	SW2-2	Gas Heat Rise (°F)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				OFF	ON	Gas Heat Rise (°C)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
		8	Dehumidification High	SW1-5	SW1-6	CFM	1122	1064	1003	943	882	820	758	697	632	567
				ON	ON											
			High Stage Cooling*	SW1-1	SW1-2	BHP	0.25	0.26	0.26	0.27	0.28	0.28	0.29	0.30	0.30	0.31
				ON	ON											
						Gas Heat Rise (°F)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
						Gas Heat Rise (°C)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
		9	High Static Cooling	SW2-8		CFM	1369	1308	1255	1204	1152	1105	1052	999	909	806
				ON		BHP	0.40	0.41	0.41	0.42	0.43	0.44	0.45	0.46	0.44	0.42

Manufacturer reserves the right to change, at any time, specifications and designs without notice and without obligations.

Dry Coil Air Delivery** - Horizontal and Downflow Discharge Sizes 24-60 208/230 VAC - 1,3 Phase (Continued)

Unit Size	Heating Rise °F (°C)	Motor Speed	Allowable Functions	Motor Speed Selection		ESP (in. W.C.)										
							0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
24060	25 - 55 (14 - 31)	1	Continuous Fan*	SW2-5	SW2-6	CFM	712	625	531	440	344	208	NA	NA	NA	NA
				OFF	OFF											
			Dehumidification Low	SW1-7	SW1-8											
				OFF	OFF											
		2	Low Stage Cooling	SW1-3	SW1-4	BHP	0.09	0.10	0.10	0.10	0.11	0.11	NA	NA	NA	NA
				OFF	OFF											
			Dehumidification Low	SW1-7	SW1-8											
				ON	OFF											
		3	Low Stage Cooling	SW1-3	SW1-4	BHP	0.11	0.12	0.13	0.13	0.13	0.14	0.14	NA	NA	NA
				ON	OFF											
			Dehumidification Low	SW1-7	SW1-8											
				OFF	ON											
		4	Low Stage Cooling	SW1-3	SW1-4	BHP	0.17	0.18	0.18	0.19	0.19	0.20	0.20	0.21	0.21	0.22
				OFF	ON											
			Dehumidification Low	SW1-7	SW1-8											
				ON	ON											
		5	Low Stage Cooling*	SW1-3	SW1-4	BHP	0.11	0.11	0.12	0.12	0.13	0.13	0.13	0.13	NA	NA
				ON	ON											
24060	25 - 55 (14 - 31)	5	Dehumidification High	SW1-5	SW1-6	CFM	1122	1064	1003	943	882	820	758	697	632	567
				OFF	OFF											
			High Stage Cooling	SW1-1	SW1-2											
				OFF	OFF											
		6	Heating*	SW2-1	SW2-2	Gas Heat Rise (°F) Gas Heat Rise (°C)	40 22	42 23	45 25	47 26	51 28	54 30	NA NA	NA NA	NA NA	NA NA
				OFF	OFF											
			Dehumidification High	SW1-5	SW1-6											
				ON	OFF											
		7	High Stage Cooling	SW1-1	SW1-2	CFM	956	883	817	747	676	604	529	450	348	241
				ON	OFF											
			Heating	SW2-1	SW2-2											
				ON	OFF											
		8	Dehumidification High	SW1-5	SW1-6	Gas Heat Rise (°F) Gas Heat Rise (°C)	47 26	51 28	55 30	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA
				ON	OFF											
			High Stage Cooling	SW1-1	SW1-2											
				ON	ON											
		9	High Static Cooling	SW2-1	SW2-2	CFM	1204	1146	1088	1031	977	919	860	802	744	682
				ON	ON											
			High Static Cooling	SW1-5	SW1-6											
				ON	ON											
		9	High Static Cooling	SW1-1	SW1-2	BHP	0.29	0.30	0.31	0.31	0.32	0.33	0.34	0.34	0.35	0.36
				ON	ON											
			High Static Cooling	SW2-1	SW2-2											
				ON	ON											
		9	High Static Cooling	SW2-1	SW2-2	Gas Heat Rise (°F) Gas Heat Rise (°C)	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA
				ON	ON											
			High Static Cooling	SW1-5	SW1-6											
				ON	ON											
		9	High Static Cooling	SW1-1	SW1-2	CFM	1369	1308	1255	1204	1152	1105	1052	999	909	806
				ON	ON											
			High Static Cooling	SW2-1	SW2-2											
				ON	ON											
		9	High Static Cooling	SW2-1	SW2-2	BHP	0.40	0.41	0.41	0.42	0.43	0.44	0.45	0.46	0.44	0.42
				ON	ON											
			High Static Cooling	SW1-5	SW1-6											
				ON	ON											

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48NGU Series B Ultra Low NOx: Product Data

Manufacturer reserves the right to change, at any time, specifications and designs without notice and without obligations.

Dry Coil Air Delivery** - Horizontal and Downflow Discharge Sizes 24-60 208/230 VAC - 1,3 Phase (Continued)

Unit Size	Heating Rise °F (°C)	Motor Speed	Allowable Functions	Motor Speed Selection		ESP (in. W.C.)										
							0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
48090	35 - 65 (19 - 36)	1	Continuous Fan*	SW2-5 OFF	SW2-6 OFF	CFM	923	812	741	677	614	549	494	444	387	338
			Dehumidification Low	SW1-7 OFF	SW1-8 OFF											
			Low Stage Cooling	SW1-3 OFF	SW1-4 OFF	BHP	0.10	0.10	0.11	0.11	0.12	0.13	0.14	0.14	0.15	0.16
		2	Continuous Fan	SW2-5 ON	SW2-6 OFF	CFM	945	885	820	757	696	638	579	527	480	429
			Dehumidification Low	SW1-7 ON	SW1-8 OFF											
			Low Stage Cooling	SW1-3 ON	SW1-4 OFF	BHP	0.11	0.12	0.12	0.13	0.14	0.15	0.16	0.16	0.17	0.18
		3	Continuous Fan	SW2-5 OFF	SW2-6 ON	CFM	1102	1051	999	945	890	837	785	734	681	634
			Dehumidification Low	SW1-7 OFF	SW1-8 ON											
			Low Stage Cooling	SW1-3 OFF	SW1-4 ON	BHP	0.15	0.16	0.17	0.18	0.19	0.20	0.21	0.22	0.23	0.24
		4	Continuous Fan	SW2-5 ON	SW2-6 ON	CFM	1297	1253	1207	1163	1115	1066	1018	974	931	888
			Dehumidification Low	SW1-7 ON	SW1-8 ON											
			Low Stage Cooling*	SW1-3 ON	SW1-4 ON	BHP	0.23	0.24	0.24	0.26	0.27	0.27	0.28	0.29	0.30	0.31
48090	35 - 65 (19 - 36)	5	Dehumidification High	SW1-5 OFF	SW1-6 OFF	CFM	1383	1339	1296	1254	1209	1163	1119	1076	1033	989
			High Stage Cooling	SW1-1 OFF	SW1-2 OFF											
			Heating*	SW2-1 OFF	SW2-2 OFF	Gas Heat Rise (°F) Gas Heat Rise (°C)	48 27	49 27	51 28	53 29	55 30	57 32	59 33	62 34	64 36	NA NA
		6	Dehumidification High	SW1-5 ON	SW1-6 OFF	CFM	1550	1511	1473	1434	1399	1362	1319	1278	1238	1202
			High Stage Cooling	SW1-1 ON	SW1-2 OFF											
			Heating	SW2-1 ON	SW2-2 OFF	Gas Heat Rise (°F) Gas Heat Rise (°C)	43 24	44 25	45 25	47 26	48 27	49 27	51 28	52 29	54 30	56 31
		7	Dehumidification High	SW1-5 OFF	SW1-6 ON	CFM	1676	1638	1604	1567	1532	1497	1460	1420	1383	1345
			High Stage Cooling	SW1-1 OFF	SW1-2 ON											
						Gas Heat Rise (°F) Gas Heat Rise (°C)	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA
		8	Dehumidification High	SW1-5 ON	SW1-6 ON	CFM	1733	1697	1662	1628	1593	1606	1571	1536	1499	1464
			High Stage Cooling*	SW1-1 ON	SW1-2 ON											
						Gas Heat Rise (°F) Gas Heat Rise (°C)	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA
		9	High Static Cooling		SW2-8	CFM	1842	1810	1773	1741	1710	1678	1645	1606	1570	1532
					ON	BHP	0.57	0.58	0.60	0.61	0.62	0.64	0.65	0.66	0.67	0.69

Manufacturer reserves the right to change, at any time, specifications and designs without notice and without obligations.

Dry Coil Air Delivery** - Horizontal and Downflow Discharge Sizes 24-60 208/230 VAC - 1,3 Phase (Continued)

Unit Size	Heating Rise °F (°C)	Motor Speed	Allowable Functions	Motor Speed Selection		ESP (in. W.C.)										
							0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
60090	35 - 65 (19 - 36)	1	Continuous Fan*	SW2-5	SW2-6	CFM	923	812	741	677	614	549	494	444	387	338
				OFF	OFF											
			Dehumidification Low	SW1-7	SW1-8	BHP	0.10	0.10	0.11	0.11	0.12	0.13	0.14	0.14	0.15	0.16
				OFF	OFF											
			Low Stage Cooling	SW1-3	SW1-4	CFM	945	885	820	757	696	638	579	527	480	429
				OFF	OFF											
		2	Continuous Fan	SW2-5	SW2-6	BHP	0.11	0.12	0.12	0.13	0.14	0.15	0.16	0.16	0.17	0.18
				ON	OFF											
			Dehumidification Low	SW1-7	SW1-8	CFM	1102	1051	999	945	890	837	785	734	681	634
				ON	OFF											
			Low Stage Cooling	SW1-3	SW1-4	BHP	0.15	0.16	0.17	0.18	0.19	0.20	0.21	0.22	0.23	0.24
				ON	OFF											
		3	Continuous Fan	SW2-5	SW2-6	CFM	1297	1253	1207	1163	1115	1066	1018	974	931	888
				OFF	ON											
			Dehumidification Low	SW1-7	SW1-8	BHP	0.23	0.24	0.24	0.26	0.27	0.27	0.28	0.29	0.30	0.31
				OFF	ON											
			Low Stage Cooling*	SW1-3	SW1-4											
				ON	ON											

Dry Coil Air Delivery - Horizontal and Downflow Discharge Sizes 24-60 208/230 VAC - 1,3 Phase (Continued)**

Unit Size	Heating Rise °F (°C)	Motor Speed	Allowable Functions	Motor Speed Selection		ESP (in. W.C.)										
							0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
60090	35 - 65 (19 - 36)	5	Dehumidification High	SW1-5	SW1-6	CFM	1383	1339	1296	1254	1209	1163	1119	1076	1033	989
				OFF	OFF											
			High Stage Cooling	SW1-1	SW1-2	BHP	0.26	0.27	0.28	0.30	0.31	0.32	0.33	0.34	0.35	0.36
				OFF	OFF											
			Heating*	SW2-1	SW2-2	Gas Heat Rise (°F)	48	49	51	53	55	57	59	62	64	NA
				OFF	OFF	Gas Heat Rise (°C)	27	27	28	29	30	32	33	34	36	NA
		6	Dehumidification High	SW1-5	SW1-6	CFM	1550	1511	1473	1434	1399	1362	1319	1278	1238	1202
				ON	OFF											
			High Stage Cooling	SW1-1	SW1-2	BHP	0.36	0.37	0.38	0.39	0.40	0.41	0.42	0.44	0.45	0.46
				ON	OFF											
			Heating	SW2-1	SW2-2	Gas Heat Rise (°F)	43	44	45	46	47	49	50	52	53	55
				ON	OFF	Gas Heat Rise (°C)	24	24	25	26	26	27	28	29	30	31
		7	Dehumidification High	SW1-5	SW1-6	CFM	1818	1783	1751	1718	1685	1652	1619	1584	1548	1512
				OFF	ON											
			High Stage Cooling	SW1-1	SW1-2	BHP	0.54	0.55	0.56	0.58	0.59	0.60	0.62	0.63	0.64	0.65
				OFF	ON											
						Gas Heat Rise (°F)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
						Gas Heat Rise (°C)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
		8	Dehumidification High	SW1-5	SW1-6	CFM	1914	1881	1849	1817	1785	1755	1723	1693	1659	1618
				ON	ON											
			High Stage Cooling	SW1-1	SW1-2	BHP	0.62	0.63	0.65	0.66	0.67	0.69	0.70	0.71	0.73	0.74
				ON	ON											
						Gas Heat Rise (°F)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
						Gas Heat Rise (°C)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
		9	High Static Cooling	SW2-8		CFM	1966	1933	1903	1872	1842	1811	1782	1751	1718	1619
				ON		BHP	0.67	0.68	0.70	0.71	0.73	0.74	0.75	0.77	0.78	0.74

Shaded areas indicate speed/static combinations that are permitted for dehumidification speed

Notes:

* - Factory Set Function

** - Deduct field-supplied air filter pressure drop and wet coil pressure drop to obtain external static pressure available for ducting

"NA" = Not Allowed for particular heating speed

Wet Coil Pressure Drop (IN. W.C.)

Unit	Standard CFM (SCFM)																		
Size	500	600	700	800	900	1000	1100	1200	1300	1400	1500	1600	1700	1800	1900	2000	2100	2200	
24	0.02	0.03	0.04	0.04	0.05	0.06													
36				0.03	0.04	0.05	0.05	0.06	0.07	0.08	0.08	0.09	0.10	0.11					
48						0.03	0.04	0.04	0.05	0.06	0.06	0.07	0.08	0.09	0.10	0.11	0.12	0.13	
60						0.03	0.04	0.04	0.05	0.06	0.06	0.07	0.08	0.09	0.10	0.11	0.12	0.13	

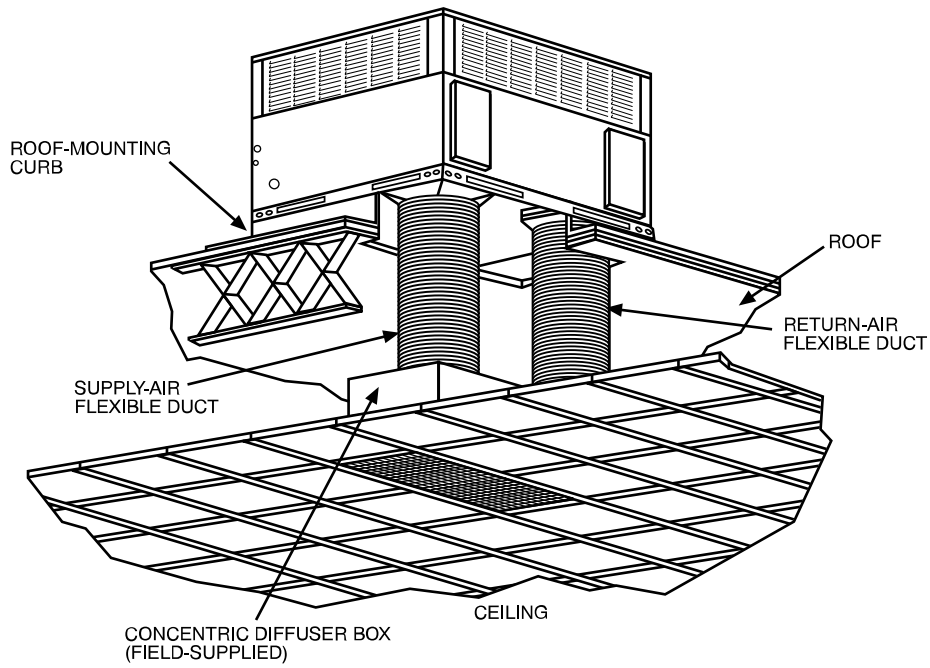
Economizer with 1-in. Filter Pressure Drop (IN. W.C.)

Filter Size in. (mm)	Cooling Tons	Standard CFM (SCFM)																	
		500	600	700	800	900	1000	1100	1200	1300	1400	1500	1600	1700	1800	1900	2000	2100	2200
600-1400 CFM 12x20x1+12x20x1 (305x508x25+305x508x25)	2.0	0.04	0.05	0.07	0.09	0.14	0.16	0.18	0.25	-	-	-	-	-	-	-	-	-	-
1200-1800 CFM 16x24x1+14x24x1 (406x610x25+356x610x25)	3.0	-	-	-	0.04	0.06	0.07	0.08	0.10	0.11	0.12	0.13	0.14	0.16	0.16	-	-	-	-
1500-2200 CFM 16x24x1+18x24x1 (406x610x25+457x610x25)	4.0	-	-	-	-	-	-	0.08	0.10	0.11	0.13	0.15	0.17	0.18	0.20	0.21	0.22	-	-
1500-2200 CFM 16x24x1+18x24x1 (406x610x25+457x610x25)	5.0	-	-	-	-	-	-	0.08	0.10	0.11	0.13	0.15	0.17	0.18	0.20	0.21	0.22	0.23	0.23

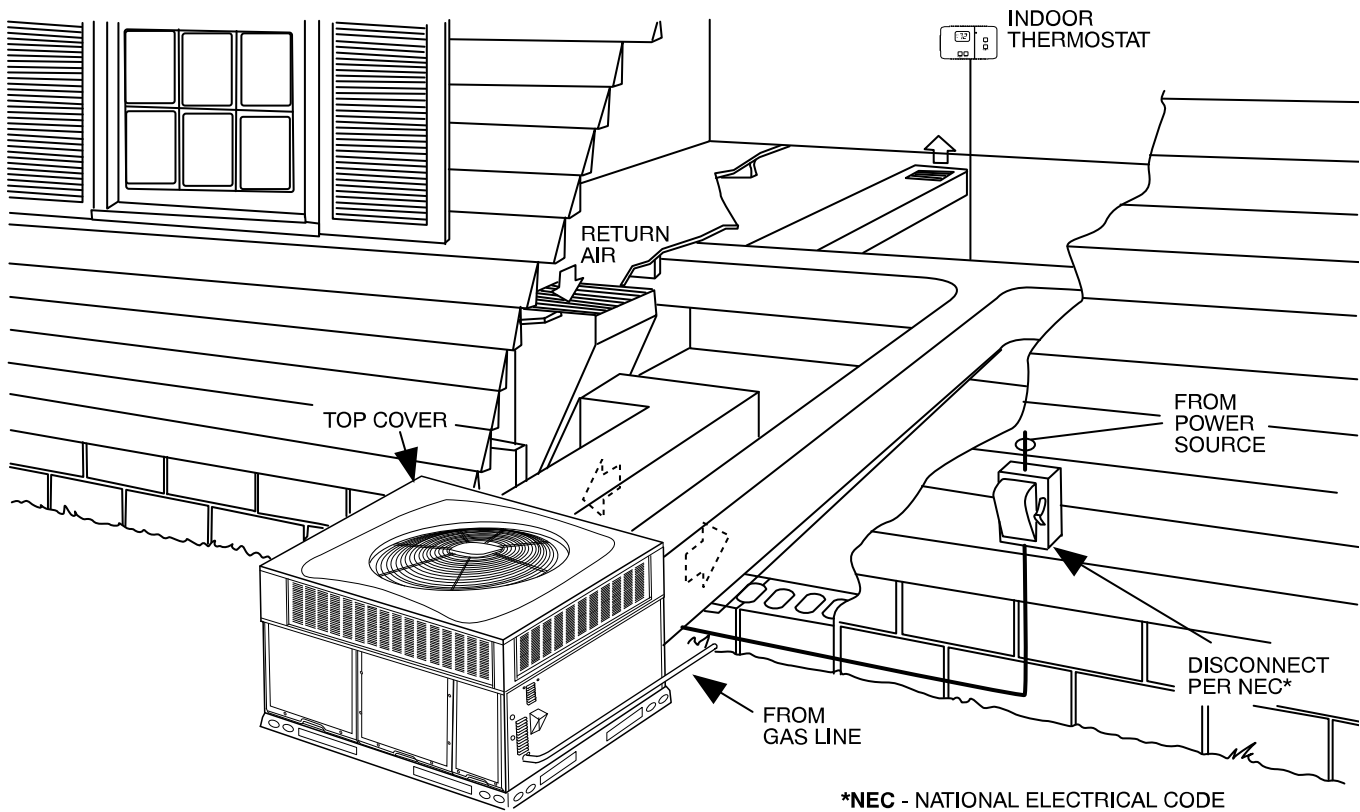
Filter Pressure Drop Table (IN. W.C.)

Filter Size in. (mm)	Cooling Tons	Standard CFM (SCFM)																	
		500	600	700	800	900	1000	1100	1200	1300	1400	1500	1600	1700	1800	1900	2000	2100	2200
600-1400 CFM 12x20x1+12x20x1 (305x508x25+305x508x25)	2.0	0.02	0.03	0.05	0.06	0.08	0.10	0.11	0.13	-	-	-	-	-	-	-	-	-	-
1200-1800 CFM 16x24x1+14x24x1 (406x610x25+356x610x25)	3.0	-	-	-	0.03	0.03	0.04	0.05	0.06	0.07	0.08	0.09	0.09	0.10	0.11	-	-	-	-
1500-2200 CFM 16x24x1+18x24x1 (406x610x25+457x610x25)	4.0	-	-	-	-	-	-	0.02	0.03	0.03	0.04	0.04	0.06	0.08	0.10	0.11	0.13	-	-
1500-2200 CFM 16x24x1+18x24x1 (406x610x25+457x610x25)	5.0	-	-	-	-	-	-	0.02	0.03	0.03	0.04	0.04	0.06	0.08	0.10	0.11	0.13	0.14	0.15

Typical Piping and Wiring



A09233



A09234

Application Data

Condensate trap — A 2-in. (50.8 mm) condensate trap must be field supplied.

Ductwork — Secure downflow discharge ductwork to roof curb. For horizontal discharge applications, attach ductwork to unit with flanges.

To convert a unit to downflow discharge — Units are equipped with factory-installed inserts in the down-flow openings. Removal of the inserts is similar to removing an electrical knock-out. Use the duct cover to seal the horizontal discharge openings in the unit. Units installed in horizontal discharge orientation do not require duct covers.

Airflow — Units are draw-thru in the cooling mode and blow-thru in the heating mode.

Electrical Data

MODEL	NOMINAL V-PH-HZ	VOLTAGE RANGE		COMPRESSOR		OFM	IFM	IDM	POWER SUPPLY	
		MIN	MAX	RLA	LRA	FLA	FLA	FLA	MCA	MOCP
24040, 24060	208/230-1-60	197	253	10.3	62	0.6	3.9	0.7	17.4	25
36060	208/230-1-60	197	253	14.4	86	1.05	5.8	0.7	24.9	35
36060	208/230-3-60	197	253	9.9	82	1.05	5.8	0.7	19.3	25
48090	208/230-1-60	197	253	19.4	102	1.05	6.9	0.7	32.3	50
48090	208/230-3-60	197	253	11.9	112	1.05	6.9	0.7	22.9	30
60090	208/230-1-60	197	253	23	148	1.05	6.9	0.7	36.8	50
60090	208/230-3-60	197	253	14	150	1.05	6.9	0.7	25.5	35

LEGEND

FLA - Full Load Amps
IDM - Inducer Motor
IFM - Indoor Fan Motor
LRA - Locked Rotor Amps
MCA - Minimum Circuit Amps
MOCP - Maximum Over Current Protection
OFM - Outdoor Fan Motor
RLA - Rated Load Amps

NOTES:

- In compliance with NEC (National Electrical Code) requirements for multimotor and combination load equipment (refer to NEC Articles 430 and 440), the overcurrent protective device for the unit shall be Power Supply fuse or circuit breaker.
- Minimum wire size is based on 60 C copper wire. If other than 60 C wire is used, or if length exceeds wire length in table, determine size from NEC.
- Unbalanced 3-Phase Supply Voltage
Never operate a motor where a phase imbalance in supply voltage is greater than 2%. Use the following formula to determine the percentage of voltage imbalance

% Voltage imbalance

$$= 100 \times \frac{\text{max voltage deviation from average voltage}}{\text{average voltage}}$$

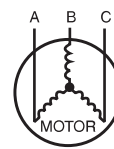
Maximum cooling airflow — To minimize the possibility of condensate blow-off from the evaporator, airflow through the units should not exceed 450 cfm per ton.

Minimum cooling airflow — Minimum cooling airflow is 350 cfm per ton.

Minimum ambient cooling operation temperature — All standard units have a minimum ambient operating temperature of 40°F (4°C). With accessory low ambient temperature kit, units can operate at temperatures down to 0°F (-17°C).

Minimum temperature — Air entering the heat exchanger in heating mode must be a minimum of 55°F (13°C) continuous and a maximum of 80°F (27°C) continuous.

EXAMPLE: Supply voltage is 230-3-60.



AB = 228 v
BC = 231 v
AC = 227 v

$$\begin{aligned} \text{Average Voltage} &= \frac{228 + 231 + 227}{3} \\ &= \frac{686}{3} \\ &= 229 \end{aligned}$$

Determine maximum deviation from average voltage.

(AB) 229 - 228 = 1 v
(BC) 231 - 229 = 2 v
(AC) 229 - 227 = 2 v

Maximum deviation is 2 v.

Determine percent of voltage imbalance

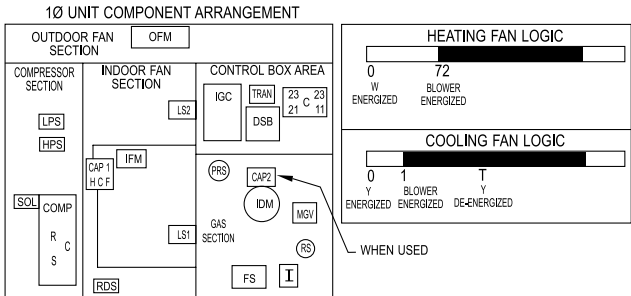
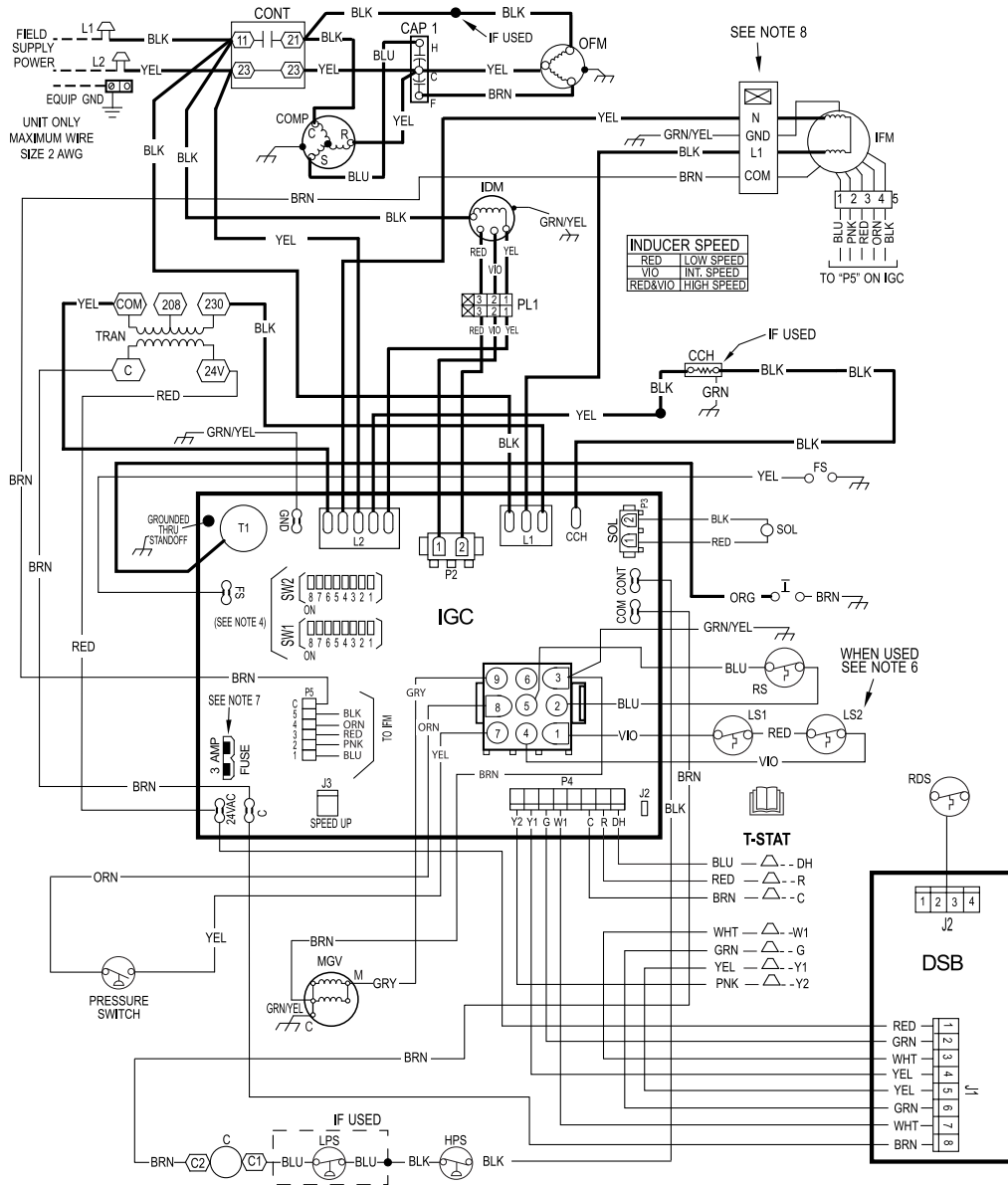
$$\begin{aligned} \% \text{ Voltage Imbalance} &= 100 \times \frac{2}{229} \\ &= 0.8\% \end{aligned}$$

This amount of phase imbalance is satisfactory as it is below the maximum allowable 2%.

IMPORTANT: If the supply voltage phase imbalance is more than 2%, contact your local electric utility company immediately.

Connection Wiring Schematic 208/230-1-60

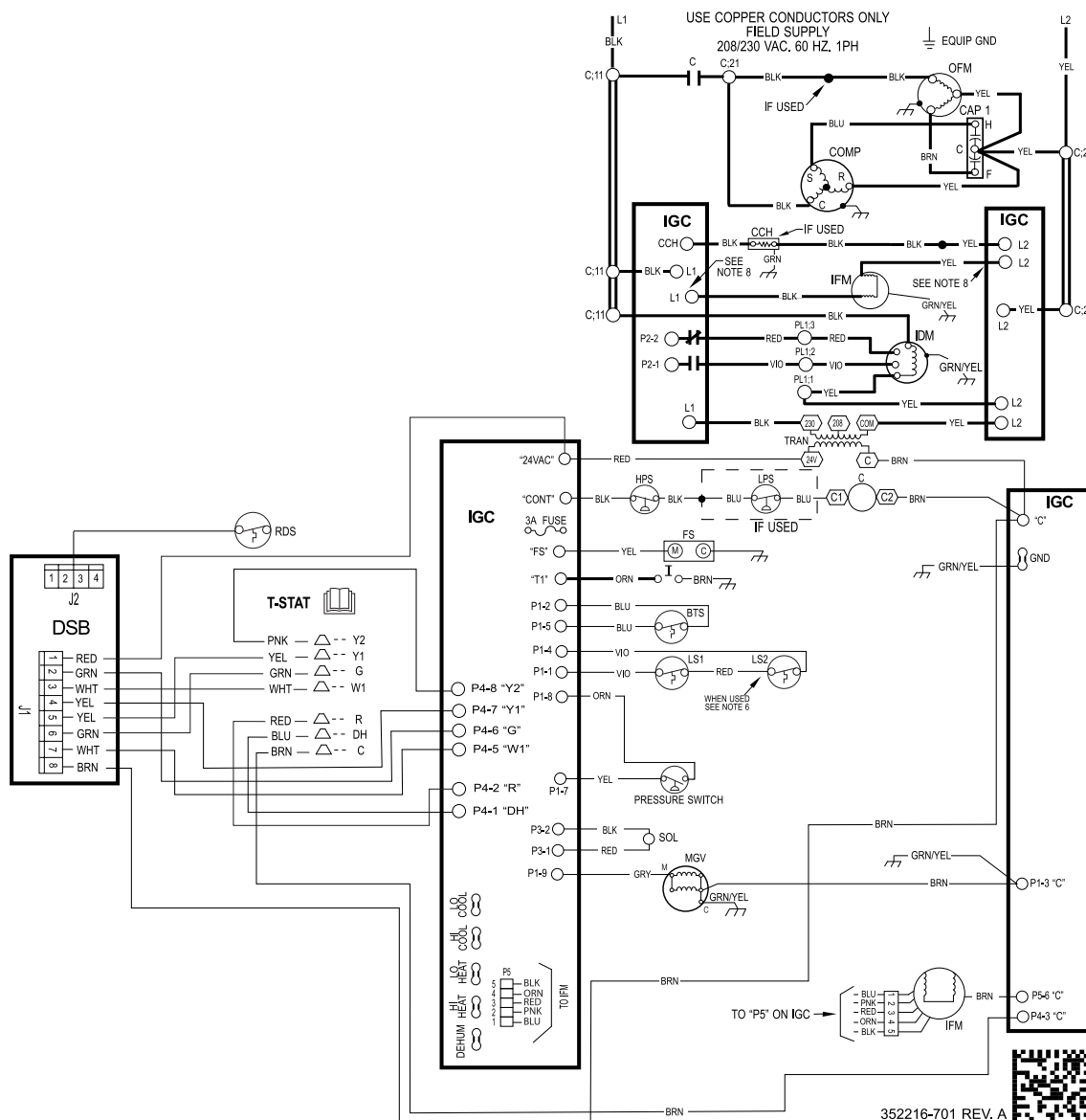
CONNECTION WIRING DIAGRAM DANGER: ELECTRICAL SHOCK HAZARD DISCONNECT POWER BEFORE SERVICING



- NOTES:**
1. IF ANY OF THE ORIGINAL WIRES FURNISHED ARE REPLACED THEY MUST BE REPLACED WITH THE SAME WIRE OR IT'S EQUIVALENT.
 2. SEE PRE-SALE LITERATURE FOR THERMOSTATS.
 3. USE 75 DEGREES C COPPER CONDUCTORS FOR FIELD INSTALLATION.
 4. REFER TO INSTALLATION INSTRUCTIONS FOR CORRECT SPEED SELECTION FOR IFM.
 5. ON SOME MODELS LS1 AND LS2 ARE WIRED IN SERIES. ON OTHER MODELS ONLY LS1 IS USED.
 6. THIS FUSE IS MANUFACTURED BY LITTLE FUSE, P/N 287003.
 7. DO NOT DISCONNECT PLUG UNDER LOAD.
 8. N.E.C. CLASS 2, 24V.
 9. INDUCER MOTOR USES A 230V CONTROL SIGNAL.

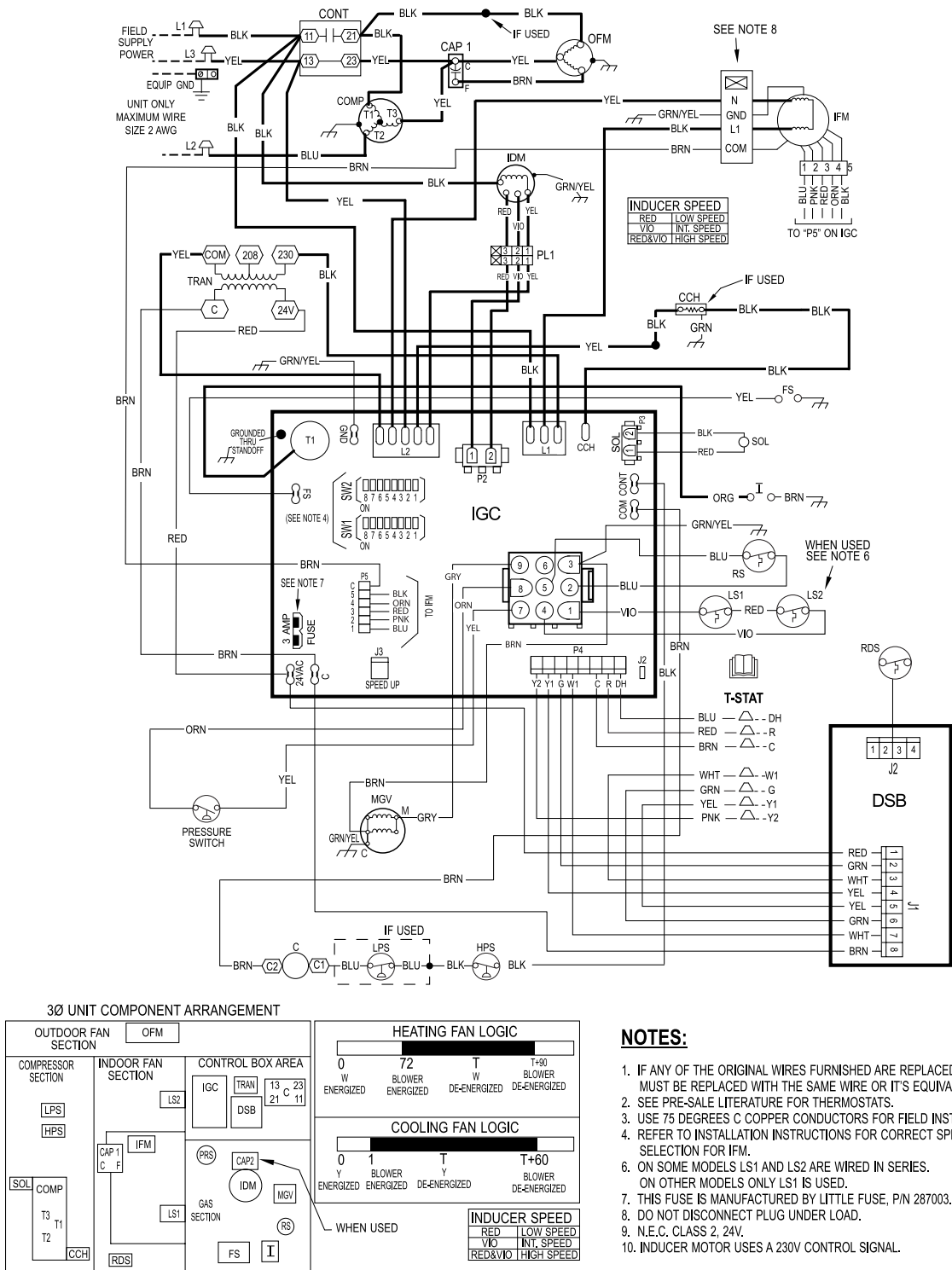
DANGER: ELECTRICAL SHOCK HAZARD DISCONNECT POWER BEFORE SERVICING

	FIELD SPICE		SPICE (MARKED)		ACCESSORY OR OPTIONAL WIRING	CAP1	CAPACITOR, COMP	CONT	COMPRESSOR CONTACTOR
	TERMINAL (MARKED)		FACTORY LOW VOLTAGE		FACTORY HI VOLTAGE	CAP2	CAPACITOR, INDUCER	DSB	DISSIPATION BOARD
	TERMINAL (UNMARKED)		FIELD CONTROL WIRING	C	CONTACTOR	CCH	CRANKCASE HEATER	EQUIP	EQUIPMENT
	SPICE		FIELD POWER WIRING	BTS	BURNER THERMAL SWITCH	COMP	COMPRESSOR MOTOR	FS	FLAME SENSOR
							PEV	PRESSURE EQUALIZER VALVE (IF USED)	
GND	GROUND	IFM	INDOOR FAN MOTOR	LS2	SECONDARY LIMIT SWITCH	RDS	REFRIG DETECTION SENSOR		
HPS	HIGH PRESSURE SWITCH	IGC	INTEGRATED GAS UNIT CONTROLLER	MGV	MAIN GAS VALVE	SOL	COMPRESSOR SOLENOID		
I	IGNITOR	LPS	LOW PRESSURE SWITCH	OFM	OUTDOOR FAN MOTOR	TRAN	TRANSFORMER		
IDM	INDUCED DRAFT MOTOR	LS1	PRIMARY LIMIT SWITCH	OT	QUADRUPLE TERMINAL	T-STAT	THERMOSTAT		



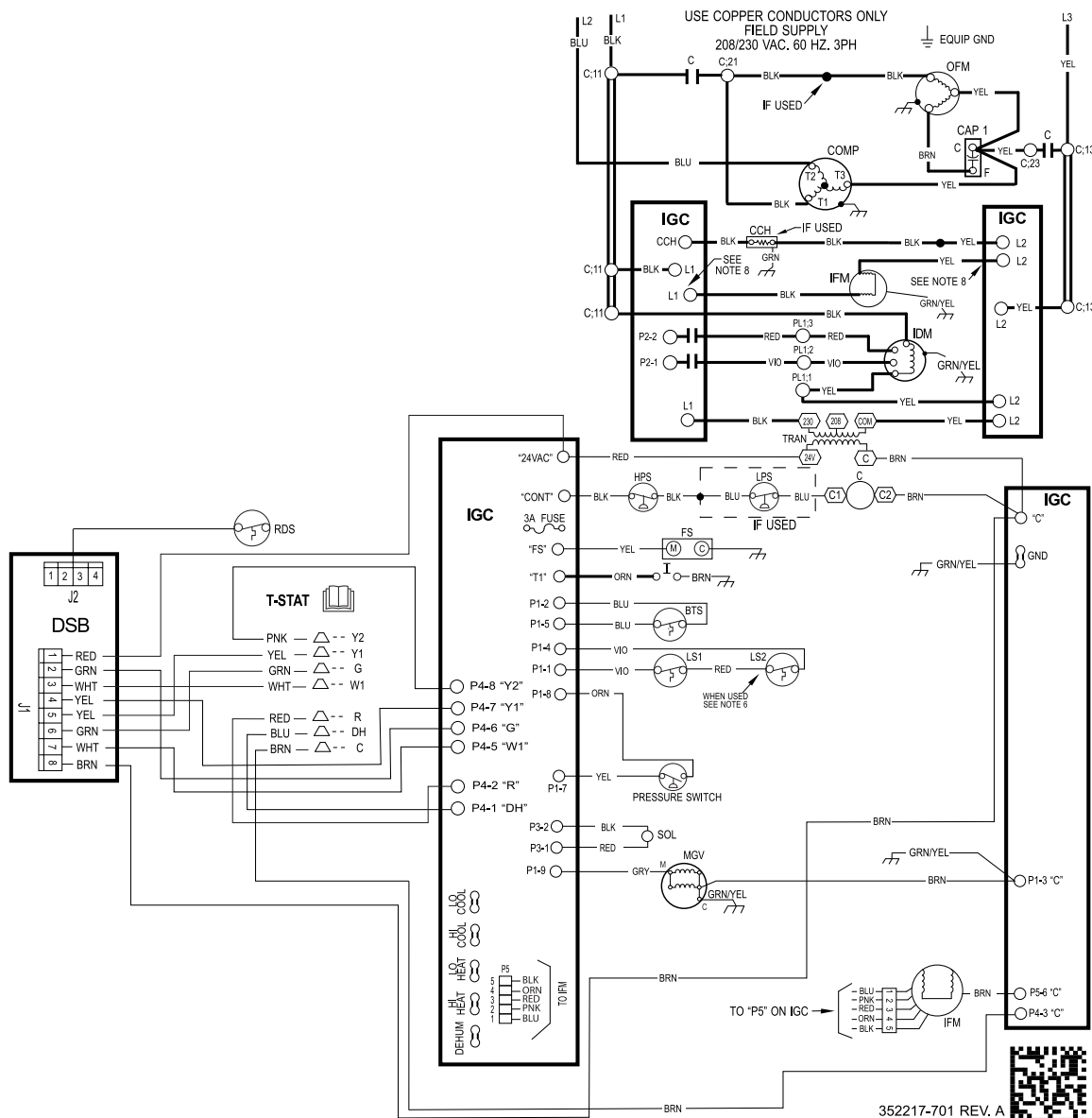
Connection Wiring Schematic Gas Inputs 40, 60, 90 KBtu/hr 208/230-3-60

CONNECTION WIRING DIAGRAM DANGER: ELECTRICAL SHOCK HAZARD DISCONNECT POWER BEFORE SERVICING



DANGER: ELECTRICAL SHOCK HAZARD DISCONNECT POWER BEFORE SERVICING

ACCESSORY OR OPTIONAL WIRING	CAP1 CAPACITOR, COMP	CONT COMPRESSOR CONTACTOR	GND GROUND
FACTORY HI VOLTAGE	CAP2 CAPACITOR, INDUCER	DSB DISSIPATION BOARD	HPS HIGH PRESSURE SWITCH
CONTACTOR	COH CRANKCASE HEATER	EQUIP EQUIPMENT	I IGNITOR
BURNER THERMAL SWITCH	COMP COMPRESSOR MOTOR	FS FLAME SENSOR	IDM INDUCED DRAFT MOTOR
		PEV PRESSURE EQUALIZER VALVE (IF USED)	
IFM INDOOR FAN MOTOR	LS2 SECONDARY LIMIT SWITCH	RDS REFRIG DETECTION SENSOR	
IGC INTEGRATED GAS UNIT CONTROLLER	MGV MAIN GAS VALVE	SOL COMPRESSOR SOLENOID	
LPS LOW PRESSURE SWITCH	OFM OUTDOOR FAN MOTOR	TRAN TRANSFORMER	
LS1 PRIMARY LIMIT SWITCH	OT QUADRUPLER TERMINAL	T-STAT THERMOSTAT	



Controls

Operating Sequence

Heating Sequence of Operation (Ultra Low NOx Single and Three Phase Models)

On a call for heating, terminal W of the thermostat is energized, starting the induced-draft motor for a 30 second pre-purge at intermediate speed. When the pressure switch senses that the induced-draft motor is moving sufficient combustion air, the burner sequence begins. This function is controlled by the integrated gas unit controller (IGC).

After 30 sec of pre-purge is complete, the pressure switch checks for sufficient combustion. Then, the gas valve energizes and the system attempts to ignite with igniter being energized. The igniter energizes for 5 sec and integrated gas controller (IGC) allows system to sense flame for 2 sec at the end of 5 sec of ignition trial. On first successful ignition, system stays at the intermediate inducer speed for 10 sec and gradually ramps up to the high speed for steady-state operation. The system uses the same logic for 2nd ignition trial. If a 3rd or 4th ignition attempt is needed, the inducer speed is decreased to the low speed. On a successful light on the 3rd or 4th attempt, the inducer speed is increased to the intermediate speed for about 10 seconds and then gradually ramps up to the high speed for steady-state operation. In case the 4th attempt fails, the system will lock out for 1 hr. After 1 hr. lock out period, the system starts with normal operating sequence. Once flame is established and proven successfully, indoor (evaporator) fan motor is energized after 30 seconds. When the thermostat is satisfied and W is de-energized, the pre-mix burner will stop firing and the indoor (evaporator) fan motor shuts off after a 90 second time-off delay. Please note that the IGC has the capability to automatically decrease the indoor (evaporator) fan motor "ON" delay and increase the indoor (evaporator) fan motor "OFF" delay in the event of high duct static and/or partially-clogged filter.

Guide Specifications

Packaged Gas Heating/Electric Cooling Units Constant Volume Application

HVAC Guide Specifications

Size Range: 2 to 5 Tons, Nominal Cooling

40,000 to 90,000 Btuh Nominal Heating Input

SYSTEM DESCRIPTION

Outdoor rooftop or ground mounted air conditioner and gas furnace system utilizing a two-stage scroll compressor for cooling duty. Unit shall discharge supply air vertically or horizontally as shown on contract drawings. Outdoor fan/coil section shall have a draw-thru design with vertical discharge for minimum sound levels.

QUALITY ASSURANCE

- Unit shall be rated in accordance with AHRI Standards 210/240 and 270-1995.
- Unit shall be designed and certified in accordance with UL Standard 60335-2-40, 60335-1 and CSA/ANSI Z21.47.
- Unit shall be manufactured in a facility registered to ISO 9001 manufacturing quality standard.
- Roof curb shall be designed to conform to NRCA Standards.
- Insulation and adhesives shall meet NFPA 90.1 requirements for flame spread and smoke generation.
- Cabinet insulation shall meet ASHRAE Standard 62.2.

DELIVERY, STORAGE AND HANDLING

Unit shall be stored and handled per manufacturer's recommendations.

Part 2 — Products

EQUIPMENT

General:

Factory-assembled, single-piece, heating and cooling unit. Contained within the enclosure shall be all factory wiring, piping, controls, refrigerant charge with R-410A refrigerant, and special features required prior to field start-up.

Unit Cabinet:

- Unit cabinet shall be constructed of phosphated, zinc-coated, pre-painted steel capable of with-standing 500 hours in salt spray.
- Normal service shall be through 3 removable cabinet panels.
- The unit shall be constructed on a rust proof unit base that has an externally trapped, integrated sloped drain.
- Evaporator fan compartment top surface shall be insulated with a minimum 1/2-in. (12.7 mm) thick, flexible fiberglass insulation, coated on the air side and retained by adhesive and mechanical means. The evaporator wall sections will be insulated with a minimum semi-rigid foil-faced board capable of being wiped clean. Aluminum foil-faced fiberglass insulation shall be used in the entire indoor air cavity section.
- Unit shall have a field-supplied condensate trap.

Fans:

- The evaporator fan shall be a multi-speed, direct-drive, as shown on equipment drawings.
- Fan wheel shall be made from steel, be double-inlet type with forward curved blades with corrosion resistant finish. Fan wheel shall be dynamically balanced.
- Condenser fan shall be direct drive propeller type with aluminum blades riveted to corrosion resistant steel spiders, be dynamically balanced, and discharge air vertically.

Compressor:

- Fully hermetic compressors with factory-installed vibration isolation.
- Two-stage scroll compressors shall be standard on all units.

Coils:

Tin-plated copper tubes on indoor coil. Copper tubes on outdoor coil.

Heating Section:

- Induced-draft combustion type with energy saving direct spark ignition system and redundant main gas valve.
- Induced-draft motors shall provide adequate airflow for combustion.
- The heat exchangers shall be constructed of stainless steel for corrosion resistance.
- Burners shall be pre-mixed type.
- All gas piping and electric power shall enter the unit cabinet at a single location.

Refrigerant Components:

Refrigerant expansion device shall be of the TXV (thermostatic expansion valve) type.

Filters:

Filter section shall consist of field-installed, throwaway, 1-in. (25 mm) thick fiberglass filters of commercially available sizes.

Controls and Safeties:

- Unit controls shall be complete with a self-contained low voltage control circuit.
- Compressors shall incorporate a solid-state compressor protector that provides reset capability.

Operating Characteristics:

- Unit shall be capable of starting and running at 125°F (51°C) ambient outdoor temperature per maximum load criteria of AHRI Standard 210.
- Compressor with standard controls shall be capable of operation down to 40°F (4°C) ambient outdoor temperature.
- Units shall be provided with fan time delay to prevent cold air delivery before the heat exchanger warms up.
- Unit shall be provided with fan time delay after the thermostat is satisfied.

Electrical Requirements:

All unit power wiring shall enter the unit cabinet at a single location.

Motors:

- Compressor motors shall be of the refrigerant-cooled type with line-break thermal and current overload protection.
- All fan motors shall have permanently lubricated bearings, and inherent, automatic reset, thermal overload protection.
- Condenser fan motor shall be totally enclosed.
- Evaporator Fan Motor to be multi-speed ECM blower motor.

Compressor Protection:

Solid-state control shall protect compressor by preventing "short cycling."

Ultra Low NOx:

Shall provide NOx reduction to meet 14ng/J NOx emissions requirements as shipped from the factory.

Guide Specifications (cont)

Accessory Kits Available:

1. Compressor Start Kit (single phase units only):
Shall provide additional starting torque for single-phase compressors.
2. Crankcase Heater Kit:
Shall provide anti-floodback protection for low-load cooling applications.
3. Economizer for two-stage operation:
(Horizontal and Vertical with Jade Honeywell W7220 controller, Honeywell communicating actuator, and dry bulb sensor. (Contact MicroMetl Customer Service at 1-800-662-4822 to order.)

NOTE: The enhanced dehumidification feature on high stage cooling does not support use of an economizer.

- a. Economizer controls capable of providing free cooling using outside air.
- b. Equipped with low leakage dampers not to exceed 3% leakage, at 1.0 IN. W.C. pressure differential.

- c. Spring return motor shuts off outdoor damper on power failure.
4. Filter Rack Kit:
Shall provide filter mounting for downflow applications. Offered as a field installed accessory.
5. Flat Roof Curb Kit:
Curbs shall have seal strip and a wood nailer for flashing and shall be installed per manufacturer's instructions.
6. Flue Discharge Deflector Kit
Directs flue gas exhaust; 90 degrees upward from current discharge.
7. Low Ambient Package Kit:
Pressure-based control to allow cooling operation down to 0°F (-17.7°C).
8. Manual Outdoor Air Damper Kit:
Package shall consist of damper, birdscreen, and rainhood which can be preset to admit outdoor air for year-round ventilation.
9. Square-To-Round Duct Transitions Kit (24-48 models):
Shall have the ability to convert the supply and return openings from rectangular to round.

Training

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